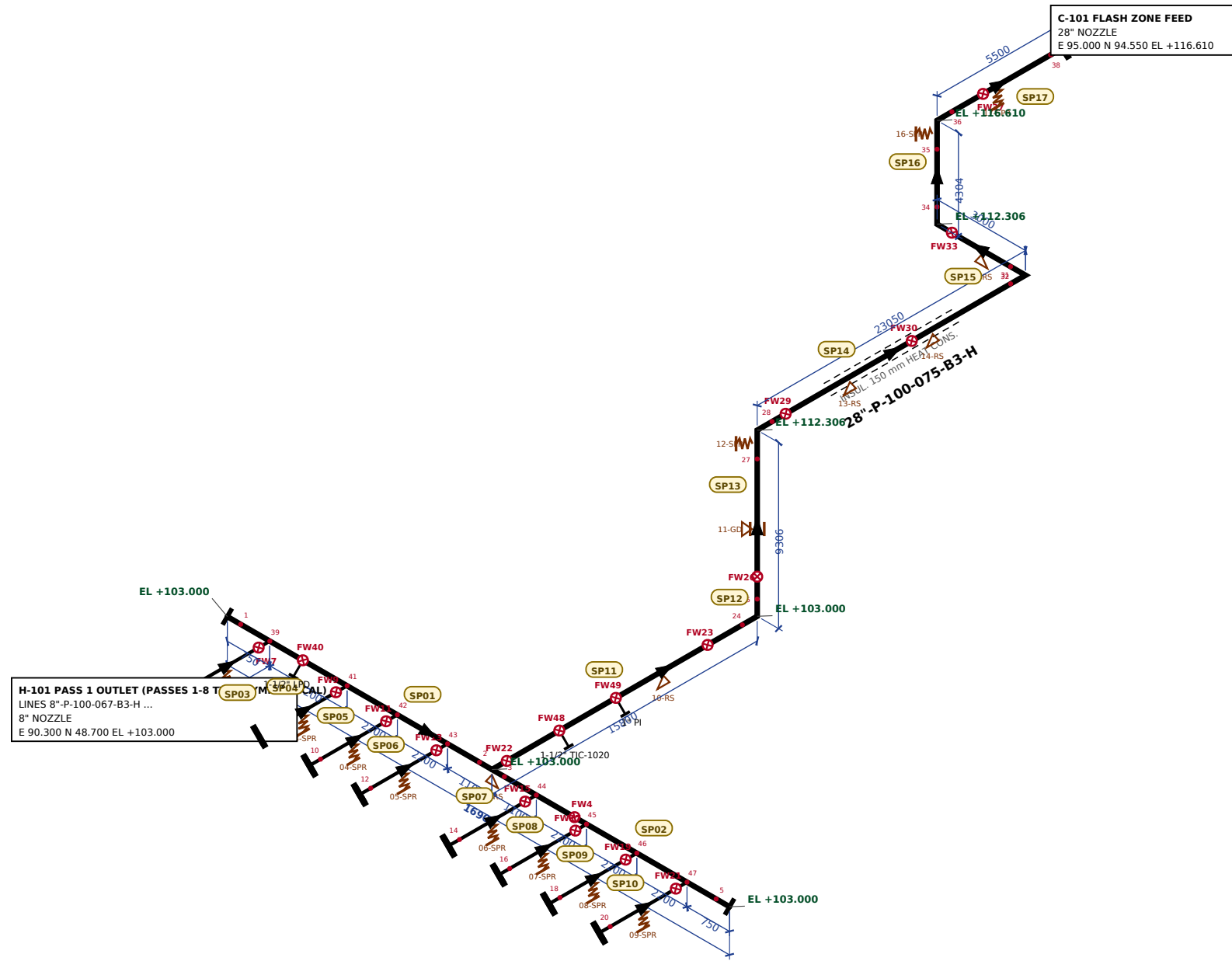
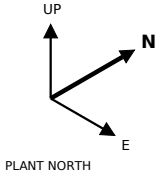


ISOMETRIC VIEW - NOT TO SCALE - DIMENSIONS IN mm, ELEVATIONS / COORDINATES IN m (PLANT GRID)



LINE NUMBER 28"-P-100-075-B3-H	FROM H-101 outlet manifold	TO C-101 flash zone	PIPING CLASS B3 - 300# 9Cr-1Mo ASTM A335 P9
DESIGN P / T 18.7 barg / 545.0 C	OPER. P / T 2.37 barg / 365.7 C	TEST P (HYDRO) 82.4 barg (1.5xPxST/S=2.94)	PIPE WALL WT 20.62 (20.62 mm)
INSULATION H 150 mm	PWHT YES - all thk (P-No.5B, B31.3 Tbl	NDE 100 % RT of BW; 100 % MT/PT of fillets;	STRESS CAT 1 - T > 300 C & NPS >= 6; T > 200 C & NPS >= 1
P&ID CFU-100-PR-PID-004, CFU-100-PR-PID-006	FLUID / PHASE P / M Atmospheric transfer lin	PAINT HT silicone/Al (insul.)	SPOOLS / WELDS 17 spools; 30 shop / 19 field

BILL OF MATERIALS (THIS SHEET)

PT	DESCRIPTION	NPS	QTY
1	CAP, BW, WT 20.62, ASTM A234 WP9	28"	2
2	PIPE WT 20.62, BE, ASTM A691 9CR Cl.42 EFW	28"	64.8 m
3	TEE EQUAL, BW, WT 20.62, ASTM A234 WP9	28"	1
4	FLANGE WN RF CL300 B16.5, bore SCH 60, ASTM A182 F9	8"	8
5	PIPE SCH 60, BE, ASTM A335 P9 SMLS	8"	10.2 m
6	ELBOW 90 LR, BW, WT 20.62, ASTM A234 WP9	28"	5
7	FLANGE WN RF CL300 B16.47A, bore WT 20.62, ASTM A182 F9	28"	1
8	WELDOLET (branch), ASTM A182 F9	8"	8
9	SOCKOLET 3000# (branch), ASTM A182 F9	1-1/2"	2
10	SOCKOLET 3000# (branch), ASTM A182 F9	1"	1
11	ERECTION: GASKET SPW 321/graphite, inner+outer ring, CL300 B16.20	8"	8
12	STUD BOLT SET (nuts) A193 B16 / A194 4, CL300	8"	8
13	GASKET SPW 321/graphite, inner+outer ring, CL300 B16.20	28"	1
14	STUD BOLT SET (nuts) A193 B16 / A194 4, CL300	28"	1
15	PIPE SUPPORTS (std. dwgs): guide x1, rest x5, spring x11	-	17

CUT LENGTHS (mm, ex. weld gap) / SPOOLS

PIECE	NPS	L	PIECE	NPS	L	PIECE	NPS	L
SP01-1	28"	7601	SP01-2	28"	2250	SP02-1	28"	5351
SP03-1	8"	1270	SP04-1	8"	1270	SP05-1	8"	1270
SP06-1	8"	1270	SP07-1	8"	1270	SP08-1	8"	1270
SP09-1	8"	1270	SP10-1	8"	1270	SP11-1	28"	11200
SP12-1	28"	3012	SP12-2	28"	1150	SP13-1	28"	6022
SP13-2	28"	1150	SP14-1	28"	11200	SP15-1	28"	8566
SP15-2	28"	866	SP16-1	28"	2171	SP16-2	28"	1150
SP17-1	28"	3083						

SPOOL DEVELOPED LENGTHS (transport limit 12.0 x 3.0 x 3.0 m):

SP01=11.2m SP02=5.7m SP03=1.4m SP04=1.4m SP05=1.4m SP06=1.4m SP07=1.4m SP08=1.4m SP09=1.4m SP10=1.4m
SP11=11.2m SP12=6.3m SP13=9.3m SP14=11.2m SP15=11.6m SP16=7.6m SP17=3.3m

NOTES

- Pipe/fittings/flanges per class B3 (CFU-000-PI-SPC-001); BW fittings ASME B16.9, flanges B16.5/B16.47A RF.
- Hydrotest 82.4 barg per ASME B31.3 345.4 (vents at high points, drains at low points).
- PWHT: YES - all thk (P-No.5B, B31.3 Tbl 331.1.1) 705-770 C. NDE: 100 % RT of BW; 100 % MT/PT of fillets; HB <= 241; 100 % PMI.
- Supports: RS rest/shoe, GD guide, ANC anchor, SPR variable spring - tags PS-<area><seq>-nn.
- Welds: dot = shop weld, crossed circle FW = field weld; spools <= 12 m for transport.
- Transfer line rises continuously from manifold to C-101 - no pockets (2-phase flow).
- Manifold symmetrical about C/L of heater: equal hydraulic length per pass.
- Rack crossing above tier 3 on dedicated transfer-line support beams at bents x=96/102.
- Formal stress analysis mandatory (API 560 terminal loads, C-101 nozzle WRC 537).

REV	DATE	DESCRIPTION
A	2026-10-02	Issued for review (FEED)

GENERIC REFINERY - GULF COAST

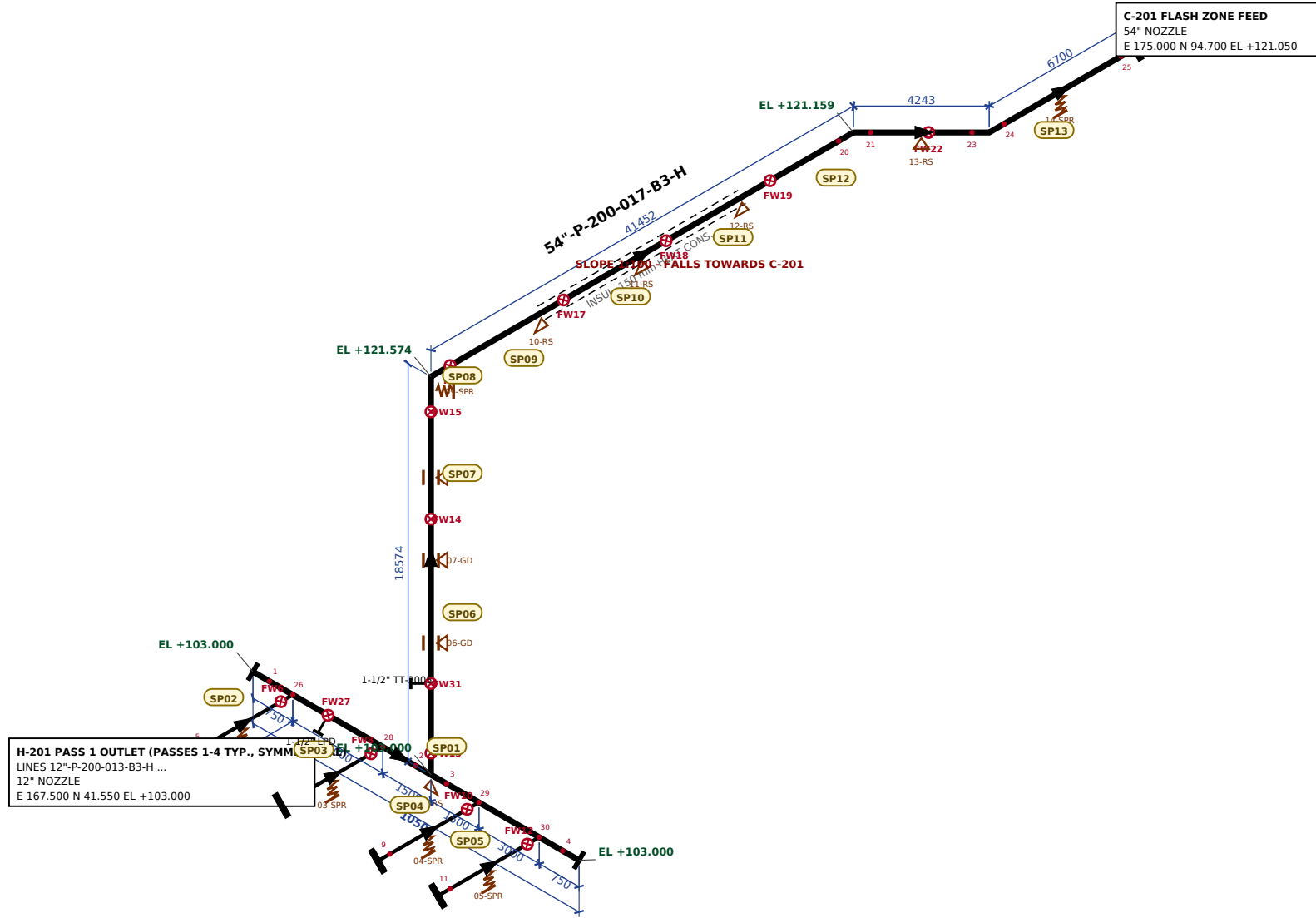
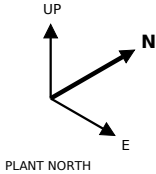
100 KBPSD CRUDE & VACUUM DISTILLATION UNIT

PIPING ISOMETRIC 28"-P-100-075-B3-H

H-101 OUTLET TRANSFER LINE TO C-101

DISCIPLINE	STAGE	SCALE	SHEET
PIPING	FEED	NTS	1 OF 1
DRAWN CLAUDE CODE	CHECKED -	APPROVED -	DATE 2026-10-02
DRAWING NUMBER CFU-100-PI-ISO-001			
REV A			

ISOMETRIC VIEW - NOT TO SCALE - DIMENSIONS IN mm, ELEVATIONS / COORDINATES IN m (PLANT GRID)



H-201 PASS 1 OUTLET (PASSES 1-4 TYP., SYMM)
LINES 12"-P-200-013-B3-H ...
12" NOZZLE
E 167.500 N 41.550 EL +103.000

C-201 FLASH ZONE FEED
54" NOZZLE
E 175.000 N 94.700 EL +121.050

BILL OF MATERIALS (THIS SHEET)

PT	DESCRIPTION	NPS	QTY
1	CAP, BW, WT 19.05, ASTM A234 WP9	54"	2
2	PIPE WT 19.05, BE, ASTM A691 9CR Cl.42 EFW	54"	69.4 m
3	TEE EQUAL, BW, WT 19.05, ASTM A234 WP9	54"	1
4	FLANGE WN RF CL300 B16.5, bore STD, ASTM A182 F9	12"	4
5	PIPE STD, BE, ASTM A335 P9 SMLS	12"	7.0 m
6	ELBOW 90 LR, BW, WT 19.05, ASTM A234 WP9	54"	1
7	ELBOW 45 LR, BW, WT 19.05, ASTM A234 WP9	54"	2
8	FLANGE WN RF CL300 B16.47A, bore WT 19.05, ASTM A182 F9	54"	1
9	WELDOLET (branch), ASTM A182 F9	12"	4
10	SOCKOLET 3000# (branch), ASTM A182 F9	1-1/2"	2
11	ERECTION: GASKET SPW 321/graphite, inner+outer ring, CL300 B16.20	12"	4
12	STUD BOLT SET (nuts) A193 B16 / A194 4, CL300	12"	4
13	GASKET SPW 321/graphite, inner+outer ring, CL300 B16.20	54"	1
14	STUD BOLT SET (nuts) A193 B16 / A194 4, CL300	54"	1
15	PIPE SUPPORTS (std. dwgs): guide x3, rest x5, spring x6	-	14

CUT LENGTHS (mm, ex. weld gap) / SPOOLS

PIECE	NPS	L	PIECE	NPS	L	PIECE	NPS	L
SP01-1	54"	3636	SP01-2	54"	3636	SP02-1	12"	1751
SP03-1	12"	1751	SP04-1	12"	1751	SP05-1	12"	1751
SP06-1	54"	10600	SP07-1	54"	4917	SP09-1	54"	10600
SP10-1	54"	10600	SP11-1	54"	10600	SP12-1	54"	6743
SP12-2	54"	1450	SP13-1	54"	1089	SP13-2	54"	5568

SPOOL DEVELOPED LENGTHS (transport limit 12.0 x 3.0 x 3.0 m):

SP01=10.5m SP02=1.9m SP03=1.9m SP04=1.9m SP05=1.9m SP06=10.6m SP07=4.9m SP08=4.1m SP09=10.6m SP10=10.6m
SP11=10.6m SP12=9.9m SP13=8.6m

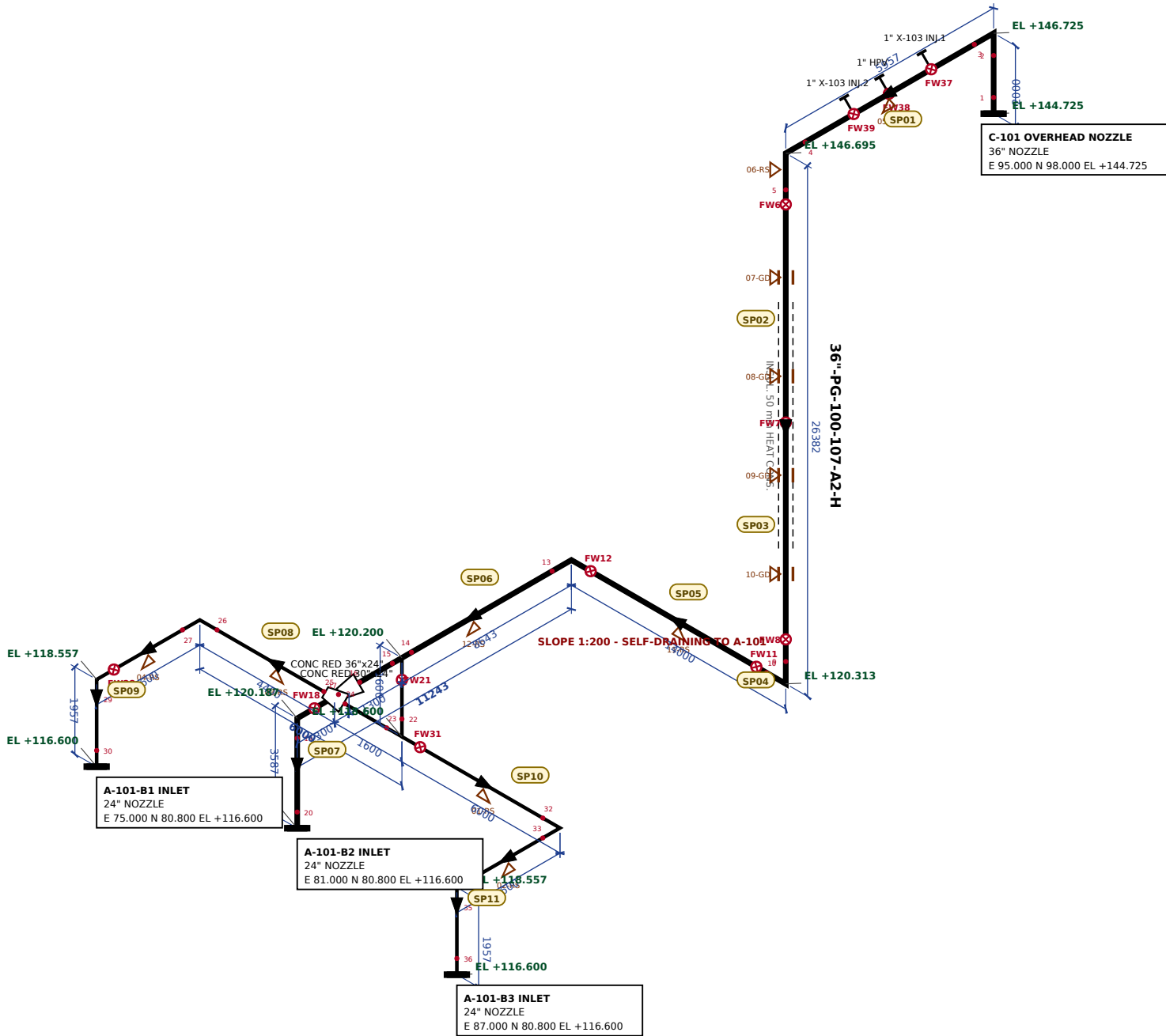
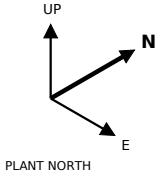
NOTES

- Pipe/fittings/flanges per class B3 (CFU-000-PI-SPC-001); BW fittings ASME B16.9, flanges B16.5/B16.47A RF.
- Hydrotest 25.2 barg per ASME B31.3 345.4 (vents at high points, drains at low points).
- PWHT: YES - all thk (P-No.5B, B31.3 Tbl 331.1.1) 705-770 C. NDE: 100 % RT of BW; 100 % MT/PT of fillets; HB <= 241; 100 % PMI.
- Supports: RS rest/shoe, GD guide, ANC anchor, SPR variable spring - tags PS-<area><seq>-nn.
- Welds: dot = shop weld, crossed circle FW = field weld; spools <= 12 m for transport.
- Vacuum transfer line: riser at heater then continuous fall 1:100 to C-201 (no liquid hold-up).
- Large bore 54" for low velocity / flash-zone pressure; external-pressure (full vacuum) design.
- Crosses rack at high level on transfer-line support structure TLS-201 (independent of rack).
- Formal stress analysis mandatory; spring hangers on riser (heater + column growth).

REV	DATE	DESCRIPTION
A	2026-10-02	Issued for review (FEED)
GENERIC REFINERY - GULF COAST 100 KBPSD CRUDE & VACUUM DISTILLATION UNIT		
PIPING ISOMETRIC 54"-P-200-017-B3-H H-201 OUTLET VACUUM TRANSFER LINE TO C-201		
DISCIPLINE PIPING	STAGE FEED	SCALE NTS
SHEET 1 OF 1		
DRAWN CLAUDE CODE	CHECKED -	APPROVED -
DATE 2026-10-02		
DRAWING NUMBER CFU-200-PI-ISO-001		REV A

LINE NUMBER 54"-P-200-017-B3-H	FROM H-201 outlet manifold	TO C-201 flash zone	PIPING CLASS B3 - 300# 9Cr-1Mo ASTM A335 P9
DESIGN P / T 3.5 barg / 580.0 C	OPER. P / T -0.7 barg / 398.4 C	TEST P (HYDRO) 25.2 barg (1.5xPxST/5=4.8)	PIPE WALL WT 19.05 (19.05 mm) ext.P
INSULATION H 150 mm	PWHT YES - all thk (P-No.5B, B31.3 Tbl	NDE 100 % RT of BW; 100 % MT/PT of fillets;	STRESS CAT 1 - T > 300 C & NPS >= 6; T > 200 C & NPS >= 1
P&ID CFU-200-PR-PID-012, CFU-200-PR-PID-013	FLUID / PHASE P / M Vacuum transfer line	PAINT HT silicone/Al (insul.)	SPOOLS / WELDS 13 spools; 17 shop / 14 field

ISOMETRIC VIEW - NOT TO SCALE - DIMENSIONS IN mm, ELEVATIONS / COORDINATES IN m (PLANT GRID)



LINE NUMBER 36"-PG-100-107-A2-H	FROM C-101 top	TO A-101	PIPING CLASS A2 - 150# CS ASTM A106 Gr.B HIC-resistan
DESIGN P / T 3.5 barg / 165.0 C	OPER. P / T 1.19 barg / 133.8 C	TEST P (HYDRO) 5.2 barg (1.5PxST/S=1.0)	PIPE WALL STD (9.53 mm)
INSULATION H 50 mm	PWHT YES - service (sour/HIC) 595-650 C	NDE 20 % RT of BW; HB <= 200	STRESS CAT 1 - air cooler header (API 661 nozzle loads);
PG&ID CFU-100-PR-PID-007, CFU-100-PR-PID-009	FLUID / PHASE PG / V C-101 overhead vapour	PAINT HT silicone/Al (insul.)	SPOOLS / WELDS 11 spools; 26 shop / 13 field

BILL OF MATERIALS (THIS SHEET)

PT	DESCRIPTION	NPS	QTY
1	FLANGE WN RF CL150 B16.47A, bore STD, ASTM A105	36"	1
2	PIPE STD, BE, ASTM A672 C70 Cl.22 EFW HIC/SSC (NACE MR0103)	36"	45.5 m
3	ELBOW 90 LR, BW, STD, ASTM A234 WPB	36"	4
4	TEE RED., BW, STD, ASTM A234 WPB	36" x 30"	1
5	REDUCER CONC, BW, STD, ASTM A234 WPB	36" x 24"	1
6	PIPE STD, BE, ASTM A106 Gr.B SMLS HIC/SSC (NACE MR0103)	24"	13.6 m
7	ELBOW 90 LR, BW, STD, ASTM A234 WPB	24"	5
8	FLANGE WN RF CL150 B16.5, bore STD, ASTM A105	24"	3
9	PIPE STD, BE, ASTM A672 C70 Cl.22 EFW HIC/SSC (NACE MR0103)	30"	1.1 m
10	TEE RED., BW, STD, ASTM A234 WPB	30" x 24"	1
11	REDUCER CONC, BW, STD, ASTM A234 WPB	30" x 24"	1
12	SOCKOLET 3000# (branch), ASTM A105	1"	3
13	ERECTION: GASKET SPW 304/graphite, outer ring, CL150 B16.20	36"	1
14	STUD BOLT SET (nuts) A193 B7 / A194 2H, CL150	36"	1
15	GASKET SPW 304/graphite, outer ring, CL150 B16.20	24"	3
16	STUD BOLT SET (nuts) A193 B7 / A194 2H, CL150	24"	3
17	PIPE SUPPORTS (std. dwgs): guide x4, rest x8	-	12

CUT LENGTHS (mm, ex. weld gap) / SPOOLS

PIECE	NPS	L	PIECE	NPS	L	PIECE	NPS	L
SP01-1	36"	472	SP01-2	36"	3214	SP01-3	36"	650
SP02-1	36"	11000	SP03-1	36"	11000	SP04-1	36"	989
SP04-2	36"	650	SP05-1	36"	10607	SP06-1	36"	6599
SP06-2	36"	322	SP06-3	24"	81	SP07-1	24"	2522
SP08-1	30"	368	SP08-2	30"	736	SP08-3	24"	3181
SP08-4	24"	771	SP09-1	24"	892	SP10-1	24"	4527
SP10-2	24"	771	SP11-1	24"	892			

SPOOL DEVELOPED LENGTHS (transport limit 12.0 x 3.0 x 3.0 m):

SP01=10.0m SP02=11.0m SP03=11.0m SP04=4.4m SP05=10.6m SP06=11.7m SP07=4.5m SP08=8.6m SP09=2.9m SP10=7.1m
SP11=2.9m

NOTES

- Pipe/fittings/flanges per class A2 (CFU-000-PI-SPC-001); BW fittings ASME B16.9, flanges B16.5/B16.47A RF.
- Hydrotest 5.2 barg per ASME B31.3 345.4 (vents at high points, drains at low points).
- PWHT: YES - service (sour/HIC) 595-650 C. NDE: 20 % RT of BW; HB <= 200.
- Supports: RS rest/shoe, GD guide, ANC anchor, SPR variable spring - tags PS-<area><seq>-nn.
- Welds: dot = shop weld, crossed circle FW = field weld; spools <= 12 m for transport.
- Line self-draining: continuous fall 1:200 from C-101 to A-101 - no pockets.
- Inlet manifold to A-101 bays symmetrical about bay B2 (hydraulic balance, NH4Cl/HCl corrosion control).
- Neutraliser / filming amine injection quills within 1.5 m of column nozzle (X-103).
- Riser guided at C-101 platform clips; weight carried by column lugs (line & column grow together).

REV	DATE	DESCRIPTION
A	2026-10-02	Issued for review (FEED)

GENERIC REFINERY - GULF COAST

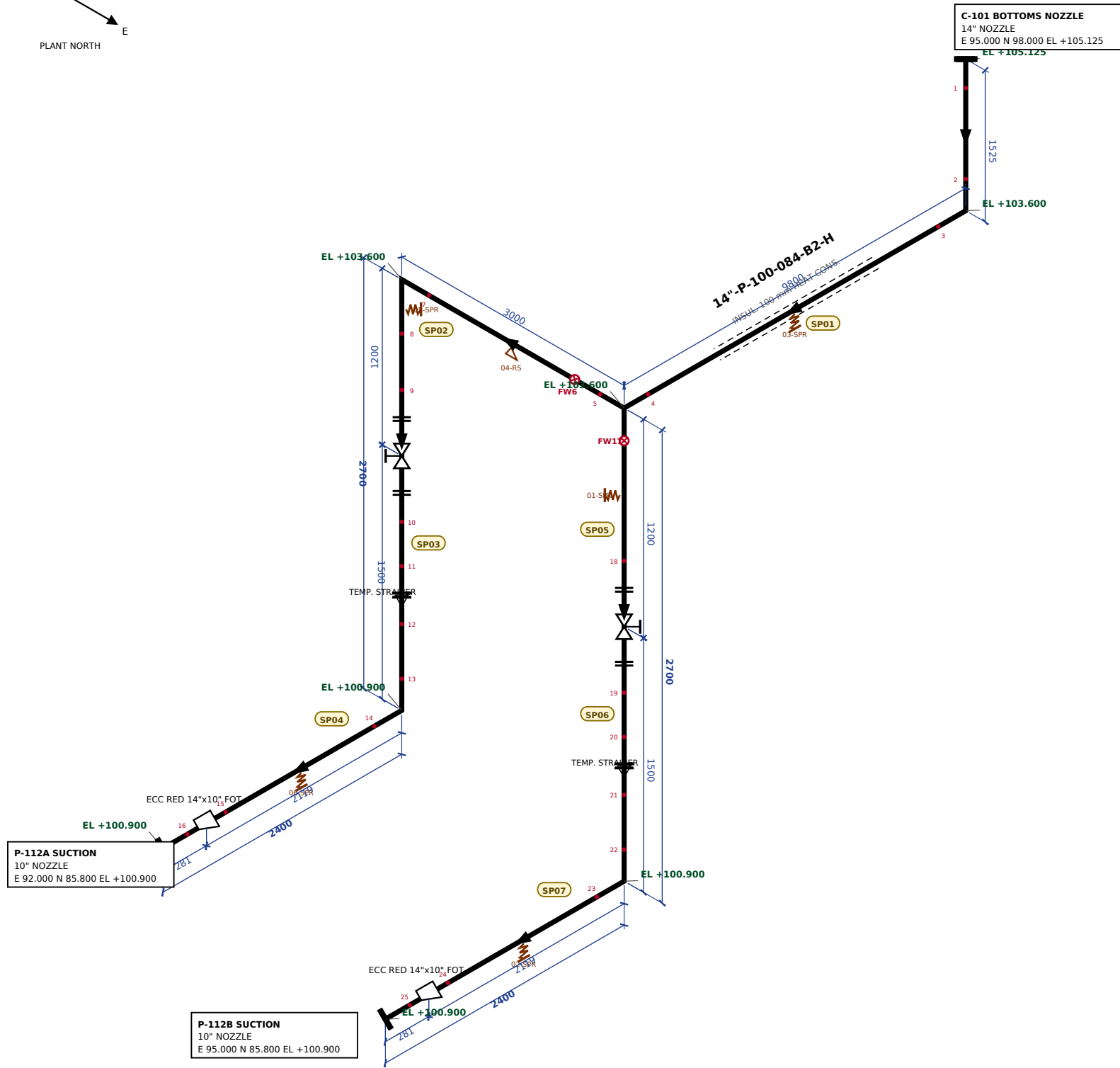
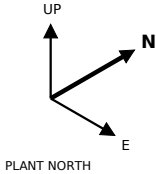
100 KBPSD CRUDE & VACUUM DISTILLATION UNIT

PIPING ISOMETRIC 36"-PG-100-107-A2-H

C-101 OVERHEAD VAPOUR TO A-101

DISCIPLINE	STAGE	SCALE	SHEET
PIPING	FEED	NTS	1 OF 1
DRAWN CLAUDE CODE	CHECKED -	APPROVED -	DATE 2026-10-02
DRAWING NUMBER CFU-100-PI-ISO-002			REV A

ISOMETRIC VIEW - NOT TO SCALE - DIMENSIONS IN mm, ELEVATIONS / COORDINATES IN m (PLANT GRID)



BILL OF MATERIALS (THIS SHEET)

PT	DESCRIPTION	NPS	QTY
1	FLANGE WN RF CL300 B16.5, bore STD, ASTM A182 F5	14"	9
2	PIPE STD, BE, ASTM A335 P5 SMLS	14"	15.7 m
3	ELBOW 90 LR, BW, STD, ASTM A234 WP5	14"	4
4	TEE EQUAL, BW, STD, ASTM A234 WP5	14"	1
5	REDUCER ECC, BW, STD, ASTM A234 WP5	14" x 10"	2
6	FLANGE WN RF CL300 B16.5, bore SCH 40, ASTM A182 F5	10"	2
7	ERECTION: GATE VALVE OS&Y CL300 RF, A217 C5, trim 5	14"	2
8	TEMPORARY CONE STRAINER (commissioning)	14"	2
9	GASKET SPW 321/graphite, inner+outer ring, CL300 B16.20	14"	9
10	STUD BOLT SET (nuts) A193 B7 / A194 4, CL300	14"	9
11	PIPE SUPPORTS (std. dwgs): rest x1, spring x5	-	6

CUT LENGTHS (mm, ex. weld gap) / SPOOLS

PIECE	NPS	L	PIECE	NPS	L	PIECE	NPS	L
SP01-1	14"	851	SP01-2	14"	8988	SP01-3	14"	350
SP02-1	14"	1838	SP02-2	14"	145	SP03-1	14"	37
SP04-1	14"	126	SP04-2	14"	1421	SP05-1	14"	399
SP06-1	14"	37	SP07-1	14"	126	SP07-2	14"	1421

SPOOL DEVELOPED LENGTHS (transport limit 12.0 x 3.0 x 3.0 m):

SP01=12.0m SP02=3.2m SP03=0.3m SP04=3.2m SP05=0.5m SP06=0.3m SP07=3.2m

NOTES

- Pipe/fittings/flanges per class B2 (CFU-000-PI-SPC-001); BW fittings ASME B16.9, flanges B16.5/B16.47A RF.
- Hydrotest 6.6 barg per ASME B31.3 345.4 (vents at high points, drains at low points).
- PWHT: YES - all thk (P-No.5B, B31.3 Tbl 331.1.1) 705-770 C. NDE: 100 % RT of BW; 100 % MT/PT of fillets; HB <= 241; 100 % PMI.
- Supports: RS rest/shoe, GD guide, ANC anchor, SPR variable spring - tags PS-<area><seq>-nn.
- Welds: dot = shop weld, crossed circle FW = field weld; spools <= 12 m for transport.
- Pump suction: no pockets, header falls to pumps; eccentric reducers 14"x10" FLAT ON TOP; temporary strain
- Pump nozzle 10" assumed (API 610 vendor data pending).
- Line exits C-101 skirt through 1000 mm access sleeve; hot (390 C) - skirt opening insulated.
- NPSHa per mech. data sheet; suction line kept short (< 15 m developed per pump).

REV	DATE	DESCRIPTION
A	2026-10-02	Issued for review (FEED)

GENERIC REFINERY - GULF COAST

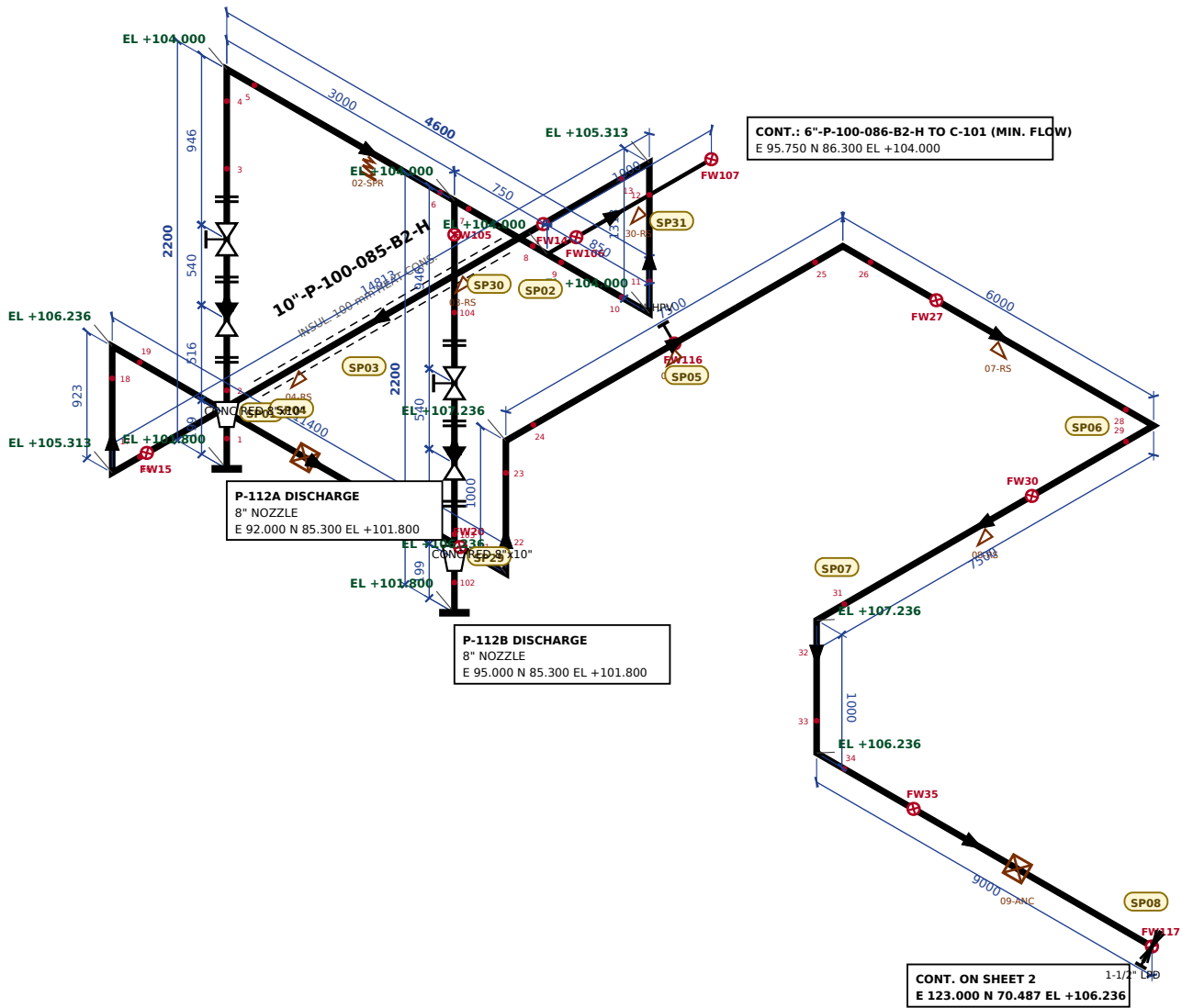
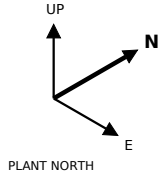
100 KBPSD CRUDE & VACUUM DISTILLATION UNIT

PIPING ISOMETRIC 14"-P-100-084-B2-H

C-101 BOTTOMS TO P-112A/B SUCTION

DISCIPLINE	STAGE	SCALE	SHEET
PIPING	FEED	NTS	1 OF 1
DRAWN	CHECKED	APPROVED	DATE
CLAUDE CODE	-	-	2026-10-02
DRAWING NUMBER	CFU-100-PI-ISO-003		
			REV
			A

ISOMETRIC VIEW - NOT TO SCALE - DIMENSIONS IN mm, ELEVATIONS / COORDINATES IN m (PLANT GRID)



BILL OF MATERIALS (THIS SHEET)

PT	DESCRIPTION	NPS	QTY
1	FLANGE WN RF CL300 B16.5, bore SCH 40, ASTM A182 F5	8"	2
2	REDUCER CONC. BW, SCH 40, ASTM A234 WP5	10" x 8"	2
3	FLANGE WN RF CL300 B16.5, bore SCH 40, ASTM A182 F5	10"	4
4	PIPE SCH 40, BE, ASTM A335 P5 SMLS	10"	62.1 m
5	ELBOW 90 LR, BW, SCH 40, ASTM A234 WP5	10"	11
6	TEE EQUAL, BW, SCH 40, ASTM A234 WP5	10"	1
7	TEE RED., BW, SCH 40, ASTM A234 WP5	10" x 6"	1
8	PIPE SCH 40, BE, ASTM A335 P5 SMLS	6"	0.8 m
9	SOCKOLET 3000# (branch), ASTM A182 F5	1"	1
10	SOCKOLET 3000# (branch), ASTM A182 F5	1-1/2"	1
11	ERECTION: CHECK VALVE SWING CL300 RF, A217 C5, trim 5	10"	2
12	GATE VALVE OS&Y CL300 RF, A217 C5, trim 5	10"	2
13	GASKET SPW 321/graphite, inner+outer ring, CL300 B16.20	10"	6
14	STUD BOLT SET (nuts) A193 B7 / A194 4, CL300	10"	6
15	PIPE SUPPORTS (std. dwgs): anchor x2, rest x6, spring x1	-	9

CUT LENGTHS (mm, ex. weld gap) / SPOOLS

PIECE	NPS	L	PIECE	NPS	L	PIECE	NPS	L
SP02-1	10"	220	SP02-2	10"	2403	SP02-3	10"	318
SP02-4	10"	253	SP02-5	10"	551	SP02-6	10"	2250
SP03-1	10"	11600	SP04-1	10"	201	SP04-2	10"	161
SP04-3	10"	10050	SP05-1	10"	588	SP05-2	10"	238
SP05-3	10"	6738	SP05-4	10"	1350	SP06-1	10"	3888
SP06-2	10"	2250	SP07-1	10"	4488	SP07-2	10"	238
SP07-3	10"	2250	SP08-1	10"	11700	SP30-1	10"	385
SP31-1	6"	784						

SPOOL DEVELOPED LENGTHS (transport limit 12.0 x 3.0 x 3.0 m):

SP01=0.4m SP02=9.3m SP03=11.6m SP04=11.9m SP05=11.2m SP06=6.9m SP07=8.5m SP08=11.7m SP29=0.4m SP30=0.5m SP31=0.8m

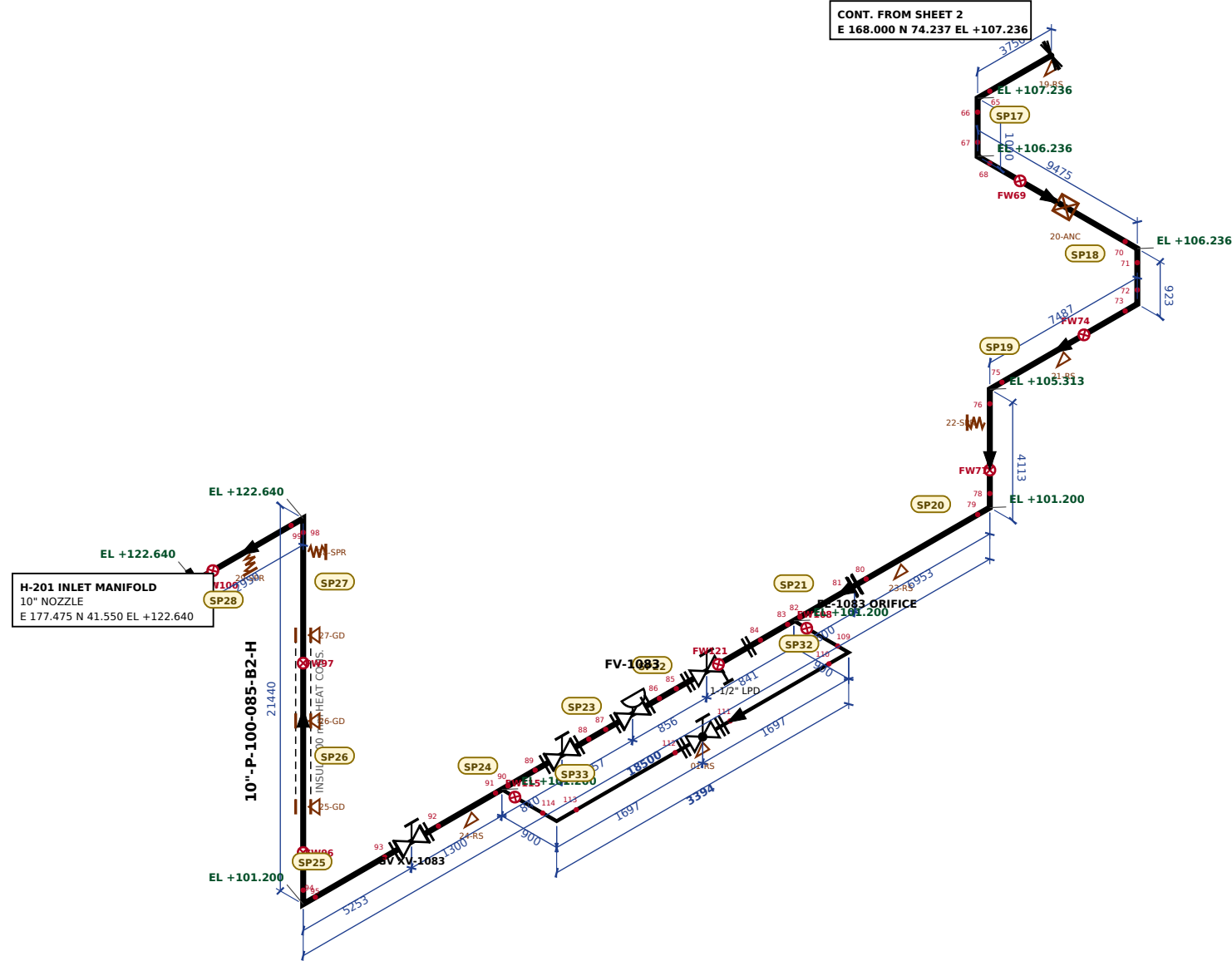
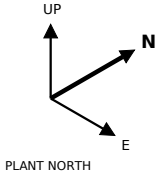
NOTES

- Pipe/fittings/flanges per class B2 (CFU-000-PI-SPC-001); BW fittings ASME B16.9, flanges B16.5/B16.47A RF.
- Hydrotest 44.7 barg per ASME B31.3 345.4 (vents at high points, drains at low points).
- PWHT: YES - all thk (P-No.5B, B31.3 Tbl 331.1.1) 705-770 C. NDE: 100 % RT of BW; 100 % MT/PT of fillets; HB <= 241; 100 % PMI.
- Supports: RS rest/shoe, GD guide, ANC anchor, SPR variable spring - tags PS-<area><seq>-nn.
- Welds: dot = shop weld, crossed circle FW = field weld; spools <= 12 m for transport.
- Discharge check + block valves in riser above each pump (accessible from grade).
- Rack run on tier 1 (hot B2) at rack south edge; expansion loops raised +1.0 m.
- FV-1083 / FE / XV station at grade (EL 101.2) north of H-201 for operator access.
- Riser at H-201 supported from heater structure with variable spring hangers.

REV	DATE	DESCRIPTION	
A	2026-10-02	Issued for review (FEED)	
GENERIC REFINERY - GULF COAST 100 KBPSD CRUDE & VACUUM DISTILLATION UNIT			
PIPING ISOMETRIC 10"-P-100-085-B2-H P-112A/B DISCHARGE TO H-201 INLET			
DISCIPLINE	STAGE	SCALE	SHEET
PIPING	FEED	NTS	1 OF 3
DRAWN	CHECKED	APPROVED	DATE
CLAUDE CODE	-	-	2026-10-02
DRAWING NUMBER			REV
CFU-100-PI-ISO-004			A

LINE NUMBER	FROM	TO	PIPING CLASS
10"-P-100-085-B2-H	P-112A/B	H-201 inlet manifold	B2 - 300# 5Cr-1/2Mo ASTM A335 P5
DESIGN P / T	OPER. P / T	TEST P (HYDRO)	PIPE WALL
24.0 barg / 380.0 C	18.99 barg / 348.7 C	44.7 barg (1.5xPxST/5=1.24)	SCH 40 (9.27 mm)
INSULATION	PWHT	NDE	STRESS
H 100 mm	YES - all thk (P-No.5B, B31.3 Tbl)	100 % RT of BW; 100 % MT/PT of fillets;	CAT 1 - T > 300 C & NPS >= 6; rotating equipment (
P&ID	FLUID / PHASE	PAINT	SPOOLS / WELDS
CFU-100-PR-PID-006, CFU-200-PR-PID-012	P / L Atmospheric residue to H	HT silicone/Al (insul.)	33 spools; 89 shop / 32 field

ISOMETRIC VIEW - NOT TO SCALE - DIMENSIONS IN mm, ELEVATIONS / COORDINATES IN m (PLANT GRID)



BILL OF MATERIALS (THIS SHEET)

PT	DESCRIPTION	NPS	QTY
1	PIPE SCH 40, BE, ASTM A335 P5 SMLS	10"	60.4 m
2	ELBOW 90 LR, BW, SCH 40, ASTM A234 WP5	10"	8
3	FLANGE WN RF CL300 B16.5, bore SCH 40, ASTM A182 F5 (orifice)	10"	2
4	TEE RED., BW, SCH 40, ASTM A234 WP5	10" x 6"	2
5	FLANGE WN RF CL300 B16.5, bore SCH 40, ASTM A182 F5	10"	9
6	PIPE SCH 40, BE, ASTM A335 P5 SMLS	6"	3.2 m
7	ELBOW 90 LR, BW, SCH 40, ASTM A234 WP5	6"	2
8	FLANGE WN RF CL300 B16.5, bore SCH 40, ASTM A182 F5	6"	2
9	SOCKOLET 3000# (branch), ASTM A182 F5	1-1/2"	1
10	ERECTION: ORIFICE PLATE + ORIFICE FLG PAIR FE-1083 (by I&C)	10"	1
11	GATE VALVE OS&Y CL300 RF, A217 C5, trim 5	10"	3
12	CONTROL VALVE FV-1083 CL300 RF (by I&C)	10"	1
13	GLOBE VALVE CL300 RF, A217 C5, trim 5	6"	1
14	GASKET SPW 321/graphite, inner+outer ring, CL300 B16.20	10"	11
15	STUD BOLT SET (nuts) A193 B7 / A194 4, CL300	10"	11
16	GASKET SPW 321/graphite, inner+outer ring, CL300 B16.20	6"	2
17	STUD BOLT SET (nuts) A193 B7 / A194 4, CL300	6"	2
18	PIPE SUPPORTS (std. dwgs): anchor x1, guide x3, rest x5, spring x3	-	12

CUT LENGTHS (mm, ex. weld gap) / SPOOLS

PIECE	NPS	L	PIECE	NPS	L	PIECE	NPS	L
SP17-1	10"	4488	SP17-2	10"	238	SP17-3	10"	2250
SP18-1	10"	6463	SP18-2	10"	161	SP18-3	10"	2250
SP19-1	10"	4475	SP19-2	10"	2250	SP20-1	10"	1102
SP20-2	10"	6456	SP21-1	10"	1268	SP21-2	10"	280
SP22-1	10"	20	SP23-1	10"	20	SP24-1	10"	280
SP24-2	10"	739	SP25-1	10"	4527	SP25-2	10"	2250
SP26-1	10"	11700	SP27-1	10"	6728	SP27-2	10"	2250
SP28-1	10"	203	SP32-1	6"	455	SP32-2	6"	1149
SP33-1	6"	1149	SP33-2	6"	455			

SPOOL DEVELOPED LENGTHS (transport limit 12.0 x 3.0 x 3.0 m):

SP17=8.5m SP18=10.4m SP19=7.5m SP20=8.4m SP21=2.2m SP22=0.3m SP23=0.3m SP24=1.7m SP25=7.7m SP26=11.7m
SP27=9.7m SP28=0.3m SP32=2.2m SP33=2.2m

NOTES

- Pipe/fittings/flanges per class B2 (CFU-000-PI-SPC-001); BW fittings ASME B16.9, flanges B16.5/B16.47A RF.
- Hydrotest 44.7 barg per ASME B31.3 345.4 (vents at high points, drains at low points).
- PWHT: YES - all thk (P-No.5B, B31.3 Tbl 331.1.1) 705-770 C. NDE: 100 % RT of BW; 100 % MT/PT of fillets; HB <= 241; 100 % PMI.
- Supports: RS rest/shoe, GD guide, ANC anchor, SPR variable spring - tags PS-<area><seq>-nn.
- Welds: dot = shop weld, crossed circle FW = field weld; spools <= 12 m for transport.
- Discharge check + block valves in riser above each pump (accessible from grade).
- Rack run on tier 1 (hot B2) at rack south edge; expansion loops raised +1.0 m.
- FV-1083 / FE / XV station at grade (EL 101.2) north of H-201 for operator access.
- Riser at H-201 supported from heater structure with variable spring hangers.

REV	DATE	DESCRIPTION
A	2026-10-02	Issued for review (FEED)

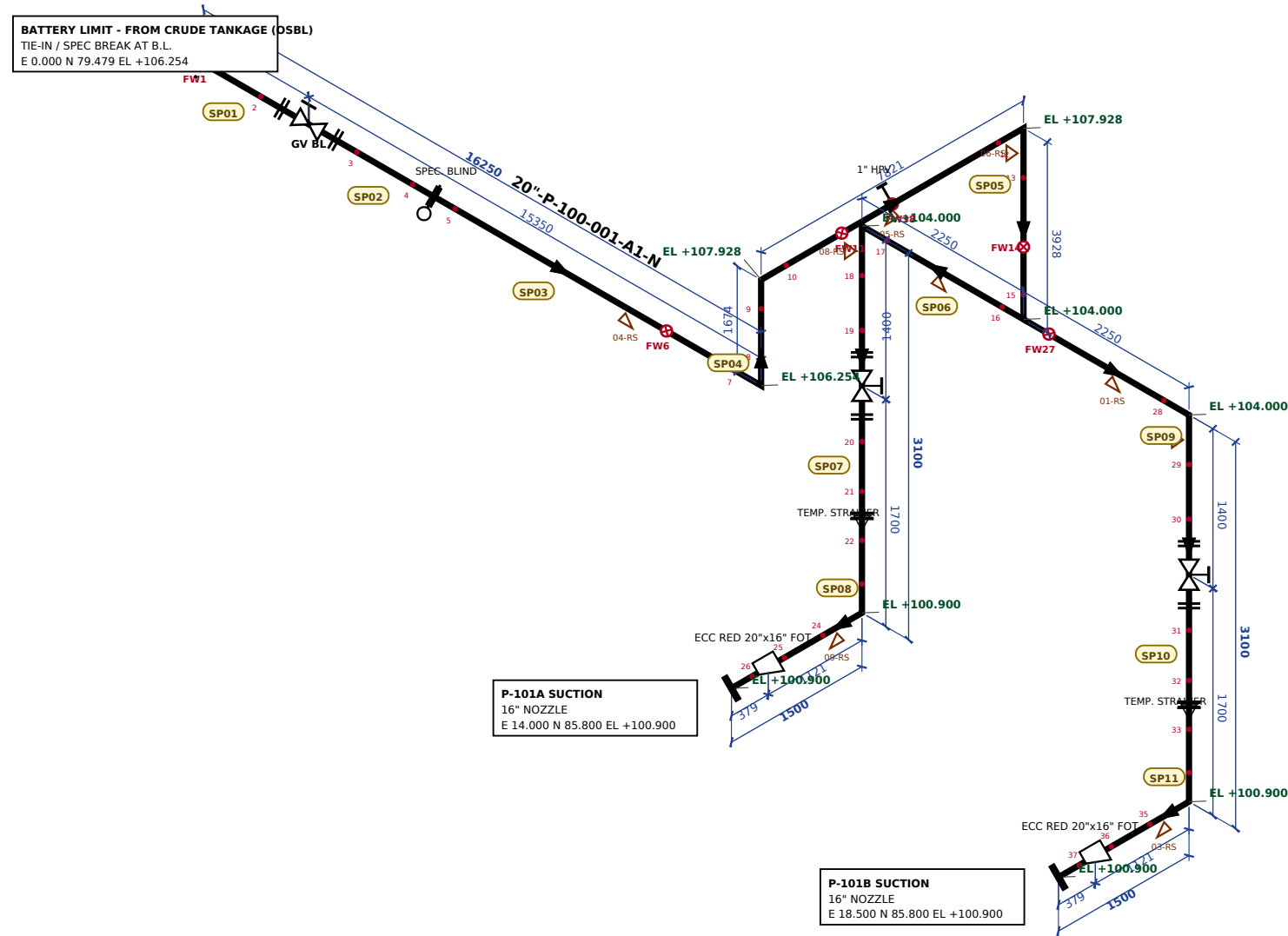
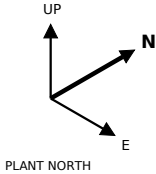
GENERIC REFINERY - GULF COAST
100 KBPSD CRUDE & VACUUM DISTILLATION UNIT

PIPING ISOMETRIC 10"-P-100-085-B2-H
P-112A/B DISCHARGE TO H-201 INLET

DISCIPLINE	STAGE	SCALE	SHEET
PIPING	FEED	NTS	3 OF 3
DRAWN	CHECKED	APPROVED	DATE
CLAUDE CODE	-	-	2026-10-02
DRAWING NUMBER	CFU-100-PI-ISO-004		
	A		

LINE NUMBER	FROM	TO	PIPING CLASS
10"-P-100-085-B2-H	P-112A/B	H-201 inlet manifold	B2 - 300# 5Cr-1/2Mo ASTM A335 P5
DESIGN P / T	OPER. P / T	TEST P (HYDRO)	PIPE WALL
24.0 barg / 380.0 C	18.99 barg / 348.7 C	44.7 barg (1.5xPxST/5=1.24)	SCH 40 (9.27 mm)
INSULATION	PWHT	NDE	STRESS
H 100 mm	YES - all thk (P-No.5B, B31.3 Tbl	100 % RT of BW; 100 % MT/PT of fillets;	CAT 1 - T > 300 C & NPS >= 6; rotating equipment (
P&ID	FLUID / PHASE	PAINT	SPOOLS / WELDS
CFU-100-PR-PID-006, CFU-200-PR-PID-012	P / L Atmospheric residue to H	HT silicone/Al (insul.)	33 spools; 89 shop / 32 field

ISOMETRIC VIEW - NOT TO SCALE - DIMENSIONS IN mm, ELEVATIONS / COORDINATES IN m (PLANT GRID)



BILL OF MATERIALS (THIS SHEET)

PT	DESCRIPTION	NPS	QTY
1	PIPE STD, BE, ASTM A106 Gr.B SMLS	20"	27.2 m
2	FLANGE WN RF CL150 B16.5, bore STD, ASTM A105	20"	12
3	ELBOW 90 LR, BW, STD, ASTM A234 WPB	20"	7
4	TEE EQUAL, BW, STD, ASTM A234 WPB	20"	1
5	REDUCER ECC, BW, STD, ASTM A234 WPB	20" x 16"	2
6	FLANGE WN RF CL150 B16.5, bore STD, ASTM A105	16"	2
7	SOCKOLET 3000# (branch), ASTM A105	1"	1
8	ERECTION: GATE VALVE OS&Y CL150 RF, A216 WCB, trim 8	20"	3
9	SPECTACLE BLIND CL150, ASTM A105	20"	1
10	TEMPORARY CONE STRAINER (commissioning)	20"	2
11	GASKET SPW 304/graphite, outer ring, CL150 B16.20	20"	12
12	STUD BOLT SET (nuts) A193 B7 / A194 2H, CL150	20"	12
13	PIPE SUPPORTS (std. dwgs): rest x9	-	9

CUT LENGTHS (mm, ex. weld gap) / SPOOLS

PIECE	NPS	L	PIECE	NPS	L	PIECE	NPS	L
SP01-1	20"	528	SP02-1	20"	485	SP03-1	20"	11250
SP04-1	20"	2195	SP04-2	20"	150	SP04-3	20"	1650
SP05-1	20"	4647	SP05-2	20"	1650	SP06-1	20"	1135
SP06-2	20"	1107	SP06-3	20"	266	SP07-1	20"	186
SP08-1	20"	95	SP08-2	20"	105	SP09-1	20"	1107
SP09-2	20"	266	SP10-1	20"	186	SP11-1	20"	95
SP11-2	20"	105						

SPOOL DEVELOPED LENGTHS (transport limit 12.0 x 3.0 x 3.0 m):

SP01=0.7m SP02=0.8m SP03=11.4m SP04=7.0m SP05=7.8m SP06=4.9m SP07=0.5m SP08=2.5m SP09=3.0m SP10=0.5m SP11=2.5m

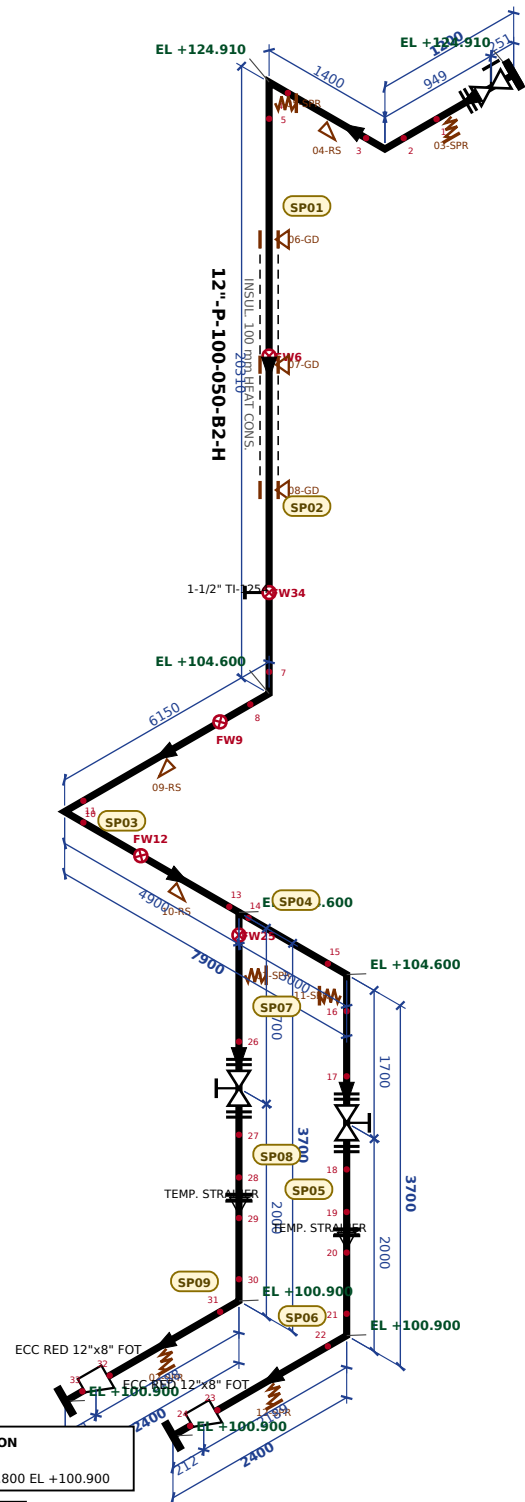
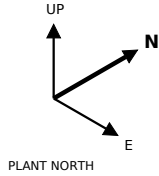
NOTES

- Pipe/fittings/flanges per class A1 (CFU-000-PI-SPC-001); BW fittings ASME B16.9, flanges B16.5/B16.47A RF.
- Hydrotest 5.6 barg per ASME B31.3 345.4 (vents at high points, drains at low points).
- PWHT: NO. NDE: 10 % RT of BW.
- Supports: RS rest/shoe, GD guide, ANC anchor, SPR variable spring - tags PS-<area><seq>-nn.
- Welds: dot = shop weld, crossed circle FW = field weld; spools <= 12 m for transport.
- Pump suction: no pockets, header falls to pumps; eccentric reducers 20"x16" FLAT ON TOP; temporary strain
- Pump nozzle 16" assumed (API 610 vendor data pending).
- Rack high point vented; tank static head gives NPSH margin (check at detailed design).
- Header at y = pump nozzle + 1.5 m limited by E-106A/B (layout) - see RPT-001 issues.

REV	DATE	DESCRIPTION
A	2026-10-02	Issued for review (FEED)
GENERIC REFINERY - GULF COAST 100 KBPSD CRUDE & VACUUM DISTILLATION UNIT		
PIPING ISOMETRIC 20"-P-100-001-A1-N CRUDE SUCTION TO P-101A/B		
DISCIPLINE PIPING	STAGE FEED	SCALE NTS
SHEET 1 OF 1		
DRAWN CLAUDE CODE	CHECKED -	APPROVED -
DATE 2026-10-02		
DRAWING NUMBER CFU-100-PI-ISO-005		REV A

LINE NUMBER 20"-P-100-001-A1-N	FROM TK (OSBL)	TO P-101A/B	PIPING CLASS A1 - 150# CS ASTM A106 Gr.B
DESIGN P / T 3.7 barg / 60.0 C	OPER. P / T 1.99 barg / 30 C	TEST P (HYDRO) 5.6 barg (1.5xPxST/S=1.0)	PIPE WALL STD (9.53 mm)
INSULATION N 0 mm	PWHT NO	NDE 10 % RT of BW	STRESS CAT 1 - rotating equipment (API 610 nozzle loads)
P&ID CFU-100-PR-PID-001	FLUID / PHASE P / L Crude from tankage to ch	PAINT Epoxy/PU system	SPOOLS / WELDS 11 spools; 32 shop / 6 field

ISOMETRIC VIEW - NOT TO SCALE - DIMENSIONS IN mm, ELEVATIONS / COORDINATES IN m (PLANT GRID)



C-101 BPA DRAW (TRAY 25)
12" NOZZLE
E 95.000 N 94.550 EL +124.910

P-108B SUCTION
8" NOZZLE
E 101.500 N 84.800 EL +100.900

P-108A SUCTION
8" NOZZLE
E 98.500 N 84.800 EL +100.900

LINE NUMBER 12"-P-100-050-B2-H	FROM C-101 tray 25	TO P-108A/B	PIPING CLASS B2 - 300# 5Cr-1/2Mo ASTM A335 P5
DESIGN P / T 3.5 barg / 345 C	OPER. P / T 1.69 barg / 313.9 C	TEST P (HYDRO) 6.4 barg (1.5xPxST/S=1.21)	PIPE WALL STD (9.53 mm)
INSULATION H 100 mm	PWHT YES - all thk (P-No.5B, B31.3 Tbl)	NDE 100 % RT of BW; 100 % MT/PT of fillets;	STRESS CAT 1 - T > 300 C & NPS >= 6; T > 200 C & NPS >= 1
P&ID CFU-100-PR-PID-007	FLUID / PHASE P / L BPA draw to pumps	PAINT HT silicone/Al (insul.)	SPOOLS / WELDS 9 spools; 29 shop / 5 field

BILL OF MATERIALS (THIS SHEET)

PT	DESCRIPTION	NPS	QTY
1	FLANGE WN RF CL300 B16.5, bore STD, ASTM A182 F5	12"	9
2	PIPE STD, BE, ASTM A335 P5 SMLS	12"	38.7 m
3	ELBOW 90 LR, BW, STD, ASTM A234 WP5	12"	7
4	TEE EQUAL, BW, STD, ASTM A234 WP5	12"	1
5	REDUCER ECC, BW, STD, ASTM A234 WP5	12" x 8"	2
6	FLANGE WN RF CL300 B16.5, bore SCH 40, ASTM A182 F5	8"	2
7	SOCKOLET 3000# (branch), ASTM A182 F5	1-1/2"	1
8	ERECTION: GATE VALVE OS&Y CL300 RF, A217 C5, trim 5	12"	3
9	TEMPORARY CONE STRAINER (commissioning)	12"	2
10	GASKET SPW 321/graphite, inner+outer ring, CL300 B16.20	12"	10
11	STUD BOLT SET (nuts) A193 B7 / A194 4, CL300	12"	10
12	PIPE SUPPORTS (std. dwgs): guide x3, rest x3, spring x6	-	12

CUT LENGTHS (mm, ex. weld gap) / SPOOLS

PIECE	NPS	L	PIECE	NPS	L	PIECE	NPS	L
SP01-1	12"	112	SP01-2	12"	486	SP01-3	12"	9350
SP02-1	12"	10046	SP02-2	12"	950	SP03-1	12"	4286
SP03-2	12"	2150	SP04-1	12"	2039	SP04-2	12"	2289
SP04-3	12"	863	SP05-1	12"	191	SP06-1	12"	714
SP06-2	12"	1630	SP07-1	12"	1066	SP08-1	12"	191
SP09-1	12"	714	SP09-2	12"	1630			

SPOOL DEVELOPED LENGTHS (transport limit 12.0 x 3.0 x 3.0 m):

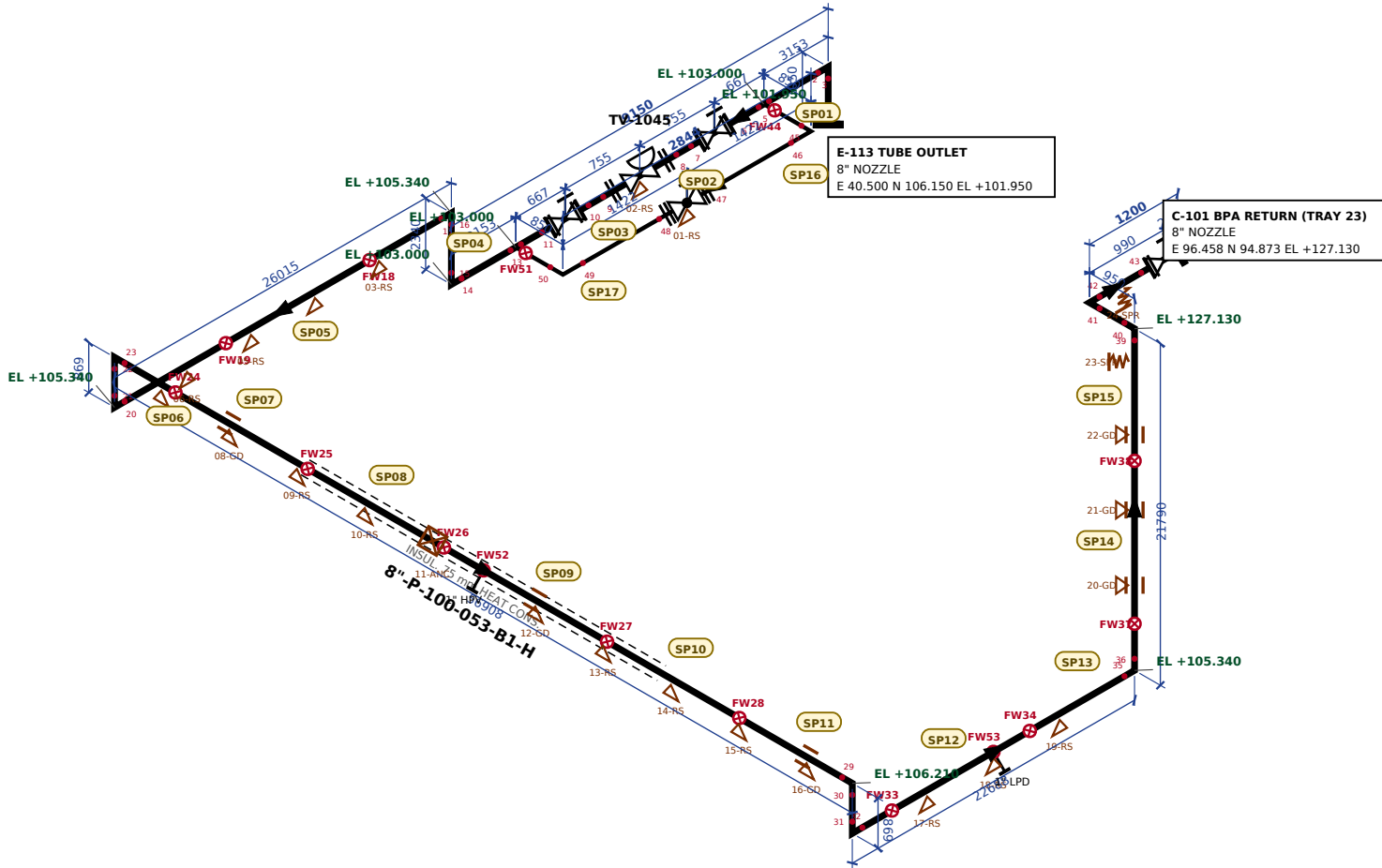
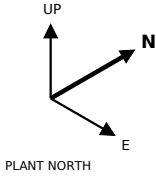
SP01=11.9m SP02=11.9m SP03=7.3m SP04=6.7m SP05=0.4m SP06=3.7m SP07=1.2m SP08=0.4m SP09=3.7m

NOTES

- Pipe/fittings/flanges per class B2 (CFU-000-PI-SPC-001); BW fittings ASME B16.9, flanges B16.5/B16.47A RF.
- Hydrotest 6.4 barg per ASME B31.3 345.4 (vents at high points, drains at low points).
- PWHT: YES - all thk (P-No.5B, B31.3 Tbl 331.1.1) 705-770 C. NDE: 100 % RT of BW; 100 % MT/PT of fillets; HB <= 241; 100 % PMI.
- Supports: RS rest/shoe, GD guide, ANC anchor, SPR variable spring - tags PS-<area><seq>-nn.
- Welds: dot = shop weld, crossed circle FW = field weld; spools <= 12 m for transport.
- Pump suction: no pockets, header falls to pumps; eccentric reducers 12"x8" FLAT ON TOP; temporary straine
- Pump nozzle 8" assumed (API 610 vendor data pending).
- Draw turns west to clear BPA return riser and C-101 OH line drop.
- Hot pump (345 C): spring support at first support off column; warm-up lines at pumps.

REV	DATE	DESCRIPTION
A	2026-10-02	Issued for review (FEED)
GENERIC REFINERY - GULF COAST 100 KBPSD CRUDE & VACUUM DISTILLATION UNIT		
PIPING ISOMETRIC 12"-P-100-050-B2-H C-101 BPA DRAW TO P-108A/B		
DISCIPLINE PIPING	STAGE FEED	SCALE NTS
SHEET 1 OF 1		
DRAWN CLAUDE CODE	CHECKED -	APPROVED -
DATE 2026-10-02		
DRAWING NUMBER CFU-100-PI-ISO-007		REV A

ISOMETRIC VIEW - NOT TO SCALE - DIMENSIONS IN mm, ELEVATIONS / COORDINATES IN m (PLANT GRID)



BILL OF MATERIALS (THIS SHEET)

PT	DESCRIPTION	NPS	QTY
1	FLANGE WN RF CL300 B16.5, bore SCH 40, ASTM A105	8"	8
2	PIPE SCH 40, BE, ASTM A106 Gr.B SMLS	8"	134.3 m
3	ELBOW 90 LR, BW, SCH 40, ASTM A234 WPB	8"	10
4	TEE RED., BW, SCH 40, ASTM A234 WPB	8" x 6"	2
5	PIPE SCH 40, BE, ASTM A106 Gr.B SMLS	6"	2.6 m
6	ELBOW 90 LR, BW, SCH 40, ASTM A234 WPB	6"	2
7	FLANGE WN RF CL300 B16.5, bore SCH 40, ASTM A105	6"	2
8	SOCKOLET 3000# (branch), ASTM A105	1"	2
9	ERECTION: GATE VALVE OS&Y CL300 RF, A216 WCB, trim 8	8"	3
10	CONTROL VALVE TV-1045 CL300 RF (by I&C)	8"	1
11	GLOBE VALVE CL300 RF, A216 WCB, trim 8	6"	1
12	GASKET SPW 304/graphite, outer ring, CL300 B16.20	8"	9
13	STUD BOLT SET (nuts) A193 B7 / A194 2H, CL300	8"	9
14	GASKET SPW 304/graphite, outer ring, CL300 B16.20	6"	2
15	STUD BOLT SET (nuts) A193 B7 / A194 2H, CL300	6"	2
16	PIPE SUPPORTS (std. dwgs): anchor x1, guide x6, rest x15, spring x2	-	24

CUT LENGTHS (mm, ex. weld gap) / SPOOLS

PIECE	NPS	L	PIECE	NPS	L	PIECE	NPS	L
SP01-1	8"	635	SP01-2	8"	2670	SP01-3	8"	170
SP02-1	8"	20	SP03-1	8"	20	SP04-1	8"	170
SP04-2	8"	2670	SP04-3	8"	1731	SP04-4	8"	5650
SP05-1	8"	11700	SP06-1	8"	8055	SP06-2	8"	260
SP06-3	8"	2450	SP07-1	8"	11700	SP08-1	8"	11700
SP09-1	8"	11700	SP10-1	8"	11700	SP11-1	8"	7048
SP11-2	8"	260	SP11-3	8"	2450	SP12-1	8"	11700
SP13-1	8"	7928	SP13-2	8"	2450	SP14-1	8"	11700
SP15-1	8"	7030	SP15-2	8"	340	SP15-3	8"	366
SP16-1	6"	443	SP16-2	6"	874	SP17-1	6"	874
SP17-2	6"	443						

SPOOL DEVELOPED LENGTHS (transport limit 12.0 x 3.0 x 3.0 m):

SP01=4.7m SP02=0.2m SP03=0.2m SP04=11.9m SP05=11.7m SP06=12.0m SP07=11.7m SP08=11.7m SP09=11.7m SP10=11.7m
SP11=11.0m SP12=11.7m SP13=11.0m SP14=11.7m SP15=9.1m SP16=1.9m SP17=1.9m

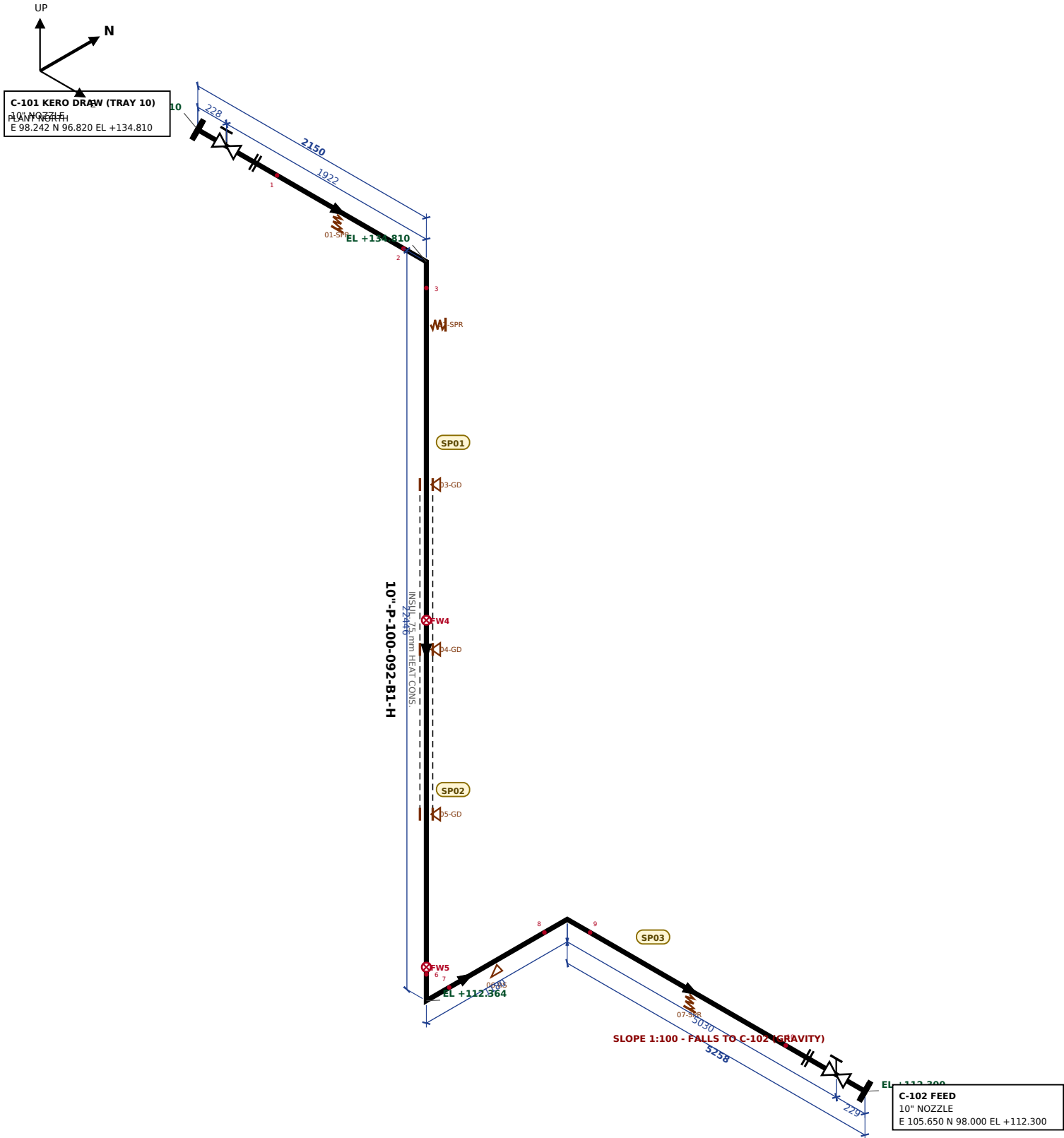
NOTES

- Pipe/fittings/flanges per class B1 (CFU-000-PI-SPC-001); BW fittings ASME B16.9, flanges B16.5/B16.47A RF.
- Hydrotest 22.5 barg per ASME B31.3 345.4 (vents at high points, drains at low points).
- PWHT: NO. NDE: 10 % RT of BW.
- Supports: RS rest/shoe, GD guide, ANC anchor, SPR variable spring - tags PS-<area><seq>-nn.
- Welds: dot = shop weld, crossed circle FW = field weld; spools <= 12 m for transport.
- TV-1045 station on platform at E-113 (EL 103.0) - access by stair from grade.
- Riser at C-101 offset 0.95 m east of nozzle to clear BPA draw and P-112 discharge crossings.

REV	DATE	DESCRIPTION
A	2026-10-02	Issued for review (FEED)
GENERIC REFINERY - GULF COAST 100 KBPSD CRUDE & VACUUM DISTILLATION UNIT		
PIPING ISOMETRIC 8"-P-100-053-B1-H BPA RETURN E-113 TO C-101		
DISCIPLINE PIPING	STAGE FEED	SCALE NTS
DRAWN CLAUDE CODE	CHECKED -	APPROVED -
DRAWING NUMBER CFU-100-PI-ISO-008	DATE 2026-10-02	REV A

LINE NUMBER 8"-P-100-053-B1-H	FROM E-113	TO C-101 tray 23	PIPING CLASS B1 - 300# CS ASTM A106 Gr.B
DESIGN P / T 13.5 barg / 290 C	OPER. P / T 3.69 barg / 223.9 C	TEST P (HYDRO) 22.5 barg (1.5xPxST/5=1.11)	PIPE WALL SCH 40 (8.18 mm)
INSULATION H 75 mm	PWHT NO	NDE 10 % RT of BW	STRESS CAT 2 - T > 150 C or equipment-connected
P&ID CFU-100-PR-PID-003, CFU-100-PR-PID-007	FLUID / PHASE P / L BPA return to column	PAINT HT silicone/Al (insul.)	SPOOLS / WELDS 17 spools; 38 shop / 15 field

ISOMETRIC VIEW - NOT TO SCALE - DIMENSIONS IN mm, ELEVATIONS / COORDINATES IN m (PLANT GRID)



LINE NUMBER 10"-P-100-092-B1-H	FROM C-101 tray 10	TO C-102	PIPING CLASS B1 - 300# CS ASTM A106 Gr.B
DESIGN P / T 3.5 barg / 250 C	OPER. P / T 1.29 barg / 219.5 C	TEST P (HYDRO) 5.5 barg (1.5xPxST/S=1.05)	PIPE WALL SCH 40 (9.27 mm)
INSULATION H 75 mm	PWHT NO	NDE 10 % RT of BW	STRESS CAT 1 - fails simplified eq.(16)
P&ID CFU-100-PR-PID-007, CFU-100-PR-PID-008	FLUID / PHASE P / L Kerosene draw to strippe	PAINT HT silicone/Al (insul.)	SPOOLS / WELDS 3 spools; 8 shop / 2 field

BILL OF MATERIALS (THIS SHEET)

PT	DESCRIPTION	NPS	QTY
1	FLANGE WN RF CL300 B16.5, bore SCH 40, ASTM A105	10"	2
2	PIPE SCH 40, BE, ASTM A106 Gr.B SMLS	10"	27.6 m
3	ELBOW 90 LR, BW, SCH 40, ASTM A234 WPB	10"	3
4	ERECTION: GATE VALVE OS&Y CL300 RF, A216 WCB, trim 8	10"	2
5	GASKET SPW 304/graphite, outer ring, CL300 B16.20	10"	4
6	STUD BOLT SET (nuts) A193 B7 / A194 2H, CL300	10"	4
7	PIPE SUPPORTS (std. dwgs): guide x3, rest x1, spring x3	-	7

CUT LENGTHS (mm, ex. weld gap) / SPOOLS

PIECE	NPS	L	PIECE	NPS	L	PIECE	NPS	L
SP01-1	10"	1196	SP01-2	10"	9850	SP02-1	10"	11600
SP03-1	10"	234	SP03-2	10"	418	SP03-3	10"	4304

SPOOL DEVELOPED LENGTHS (transport limit 12.0 x 3.0 x 3.0 m):

SP01=11.9m SP02=11.6m SP03=6.6m

NOTES

- Pipe/fittings/flanges per class B1 (CFU-000-PI-SPC-001); BW fittings ASME B16.9, flanges B16.5/B16.47A RF.
- Hydrotest 5.5 barg per ASME B31.3 345.4 (vents at high points, drains at low points).
- PWHT: NO. NDE: 10 % RT of BW.
- Supports: RS rest/shoe, GD guide, ANC anchor, SPR variable spring - tags PS-<area><seq>-nn.
- Welds: dot = shop weld, crossed circle FW = field weld; spools <= 12 m for transport.
- Gravity draw: vertical drop then continuous fall 1:100 to C-102 - no pockets, no rise.
- Drop leg 2.15 m east of draw nozzle clears diesel/AGO draw nozzles below (same orientation).
- Self-venting: line sized for <= 0.9 m/s; vapour vents back to C-101 via 8"-PG-100-093.

REV	DATE	DESCRIPTION
A	2026-10-02	Issued for review (FEED)

GENERIC REFINERY - GULF COAST

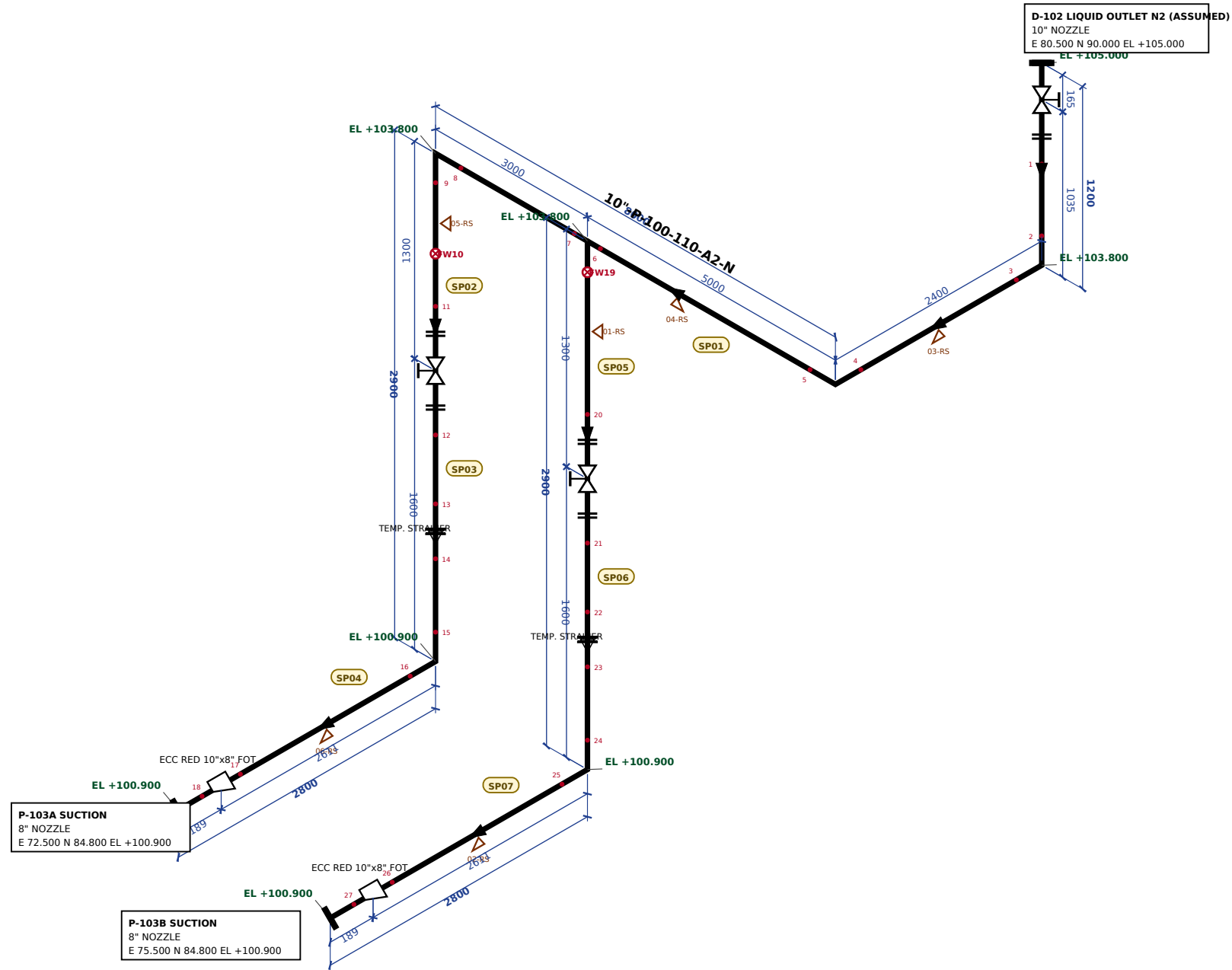
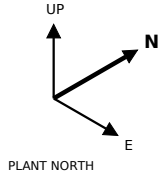
100 KBPSD CRUDE & VACUUM DISTILLATION UNIT

PIPING ISOMETRIC 10"-P-100-092-B1-H

C-101 KERO DRAW TO C-102 (GRAVITY)

DISCIPLINE PIPING	STAGE FEED	SCALE NTS	SHEET 1 OF 1
DRAWN CLAUDE CODE	CHECKED -	APPROVED -	DATE 2026-10-02
DRAWING NUMBER CFU-100-PI-ISO-009			REV A

ISOMETRIC VIEW - NOT TO SCALE - DIMENSIONS IN mm, ELEVATIONS / COORDINATES IN m (PLANT GRID)



BILL OF MATERIALS (THIS SHEET)

PT	DESCRIPTION	NPS	QTY
1	FLANGE WN RF CL150 B16.5, bore SCH 40, ASTM A105	10"	9
2	PIPE SCH 40, BE, ASTM A106 Gr.B SMLS HIC/SSC (NACE MR0103)	10"	16.1 m
3	ELBOW 90 LR, BW, SCH 40, ASTM A234 WPB	10"	5
4	TEE EQUAL, BW, SCH 40, ASTM A234 WPB	10"	1
5	REDUCER ECC, BW, SCH 40, ASTM A234 WPB	10" x 8"	2
6	FLANGE WN RF CL150 B16.5, bore SCH 40, ASTM A105	8"	2
7	ERECTION: GATE VALVE OS&Y CL150 RF, A216 WCB, trim 8	10"	3
8	TEMPORARY CONE STRAINER (commissioning)	10"	2
9	GASKET SPW 304/graphite, outer ring, CL150 B16.20	10"	10
10	STUD BOLT SET (nuts) A193 B7 / A194 2H, CL150	10"	10
11	PIPE SUPPORTS (std. dwgs): rest x6	-	6

CUT LENGTHS (mm, ex. weld gap) / SPOOLS

PIECE	NPS	L	PIECE	NPS	L	PIECE	NPS	L
SP01-1	10"	389	SP01-2	10"	1638	SP01-3	10"	4403
SP01-4	10"	2403	SP01-5	10"	250	SP02-1	10"	404
SP03-1	10"	335	SP04-1	10"	419	SP04-2	10"	2141
SP05-1	10"	819	SP06-1	10"	335	SP07-1	10"	419
SP07-2	10"	2141						

SPOOL DEVELOPED LENGTHS (transport limit 12.0 x 3.0 x 3.0 m):

SP01=11.9m SP02=0.5m SP03=0.5m SP04=3.7m SP05=0.9m SP06=0.5m SP07=3.7m

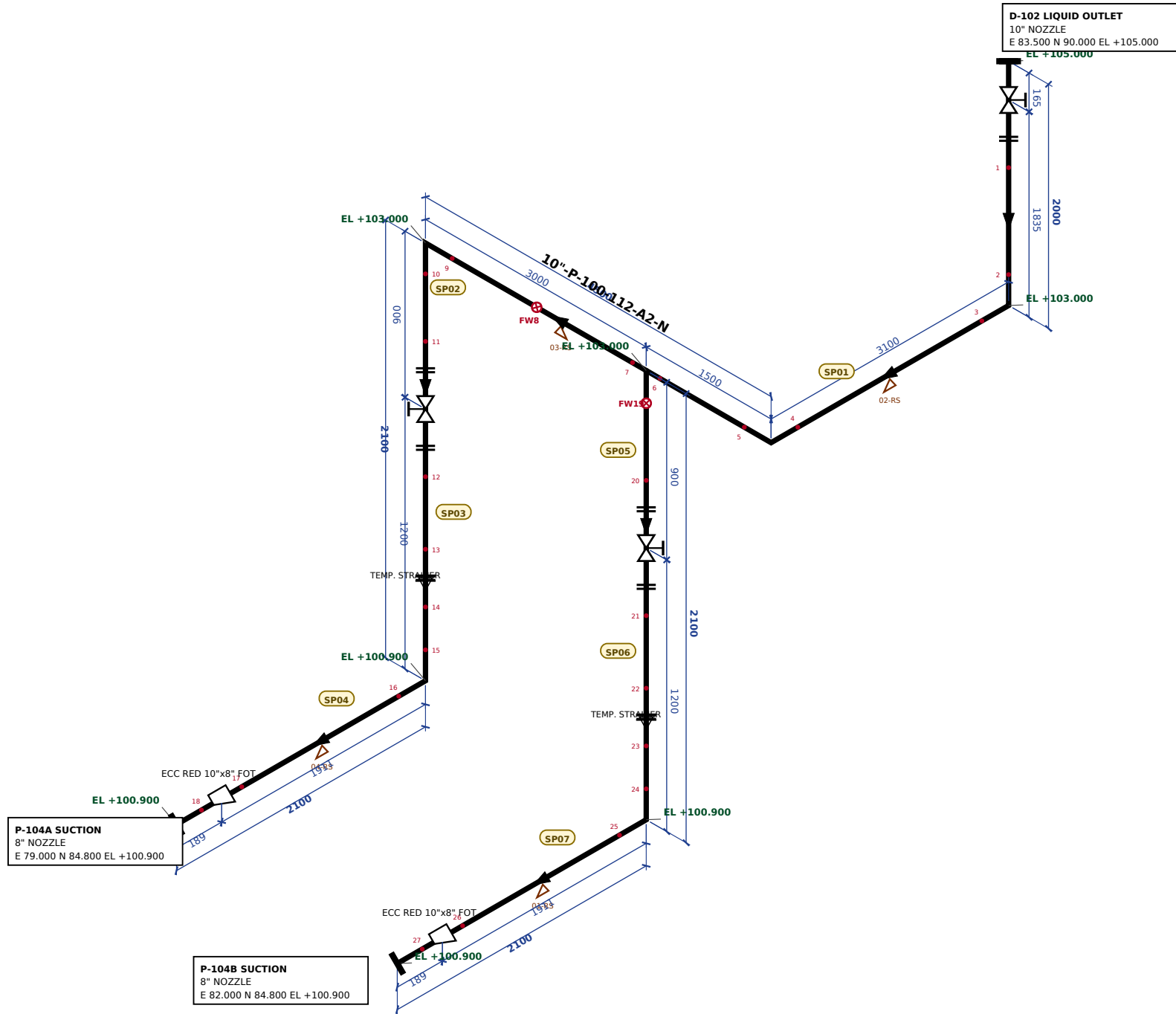
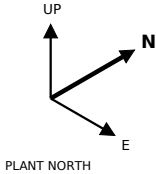
NOTES

- Pipe/fittings/flanges per class A2 (CFU-000-PI-SPC-001); BW fittings ASME B16.9, flanges B16.5/B16.47A RF.
- Hydrotest 5.2 barg per ASME B31.3 345.4 (vents at high points, drains at low points).
- PWHT: YES - service (sour/HIC) 595-650 C. NDE: 20 % RT of BW; HB <= 200.
- Supports: RS rest/shoe, GD guide, ANC anchor, SPR variable spring - tags PS-<area><seq>-nn.
- Welds: dot = shop weld, crossed circle FW = field weld; spools <= 12 m for transport.
- Pump suction: no pockets, header falls to pumps; eccentric reducers 10"x8" FLAT ON TOP; temporary straine
- Pump nozzle 8" assumed (API 610 vendor data pending).
- Vortex breaker in D-102 outlet nozzle (by vessel).
- Common header, equal branch to each pump.

REV	DATE	DESCRIPTION
A	2026-10-02	Issued for review (FEED)
GENERIC REFINERY - GULF COAST 100 KBPSD CRUDE & VACUUM DISTILLATION UNIT		
PIPING ISOMETRIC 10"-P-100-110-A2-N D-102 TO P-103A/B SUCTION		
DISCIPLINE PIPING	STAGE FEED	SCALE NTS
SHEET 1 OF 1		
DRAWN CLAUDE CODE	CHECKED -	APPROVED -
DATE 2026-10-02		
DRAWING NUMBER CFU-100-PI-ISO-010		REV A

LINE NUMBER	FROM	TO	PIPING CLASS
10"-P-100-110-A2-N	D-102	P-103A/B	A2 - 150# CS ASTM A106 Gr.B HIC-resistan
DESIGN P / T 3.5 barg / 75 C	OPER. P / T 0.69 barg / 45.0 C	TEST P (HYDRO) 5.2 barg (1.5xPxST/S=1.0)	PIPE WALL SCH 40 (9.27 mm)
INSULATION N 0 mm	PWHT YES - service (sour/HIC) 595-650 C	NDE 20 % RT of BW; HB <= 200	STRESS CAT 2 - T > 150 C or equipment-connected
P&ID CFU-100-PR-PID-009	FLUID / PHASE P / L Reflux pump suction	PAINT Epoxy/PU system	SPOOLS / WELDS 7 spools; 25 shop / 2 field

ISOMETRIC VIEW - NOT TO SCALE - DIMENSIONS IN mm, ELEVATIONS / COORDINATES IN m (PLANT GRID)



LINE NUMBER 10"-P-100-112-A2-N	FROM D-102	TO P-104A/B	PIPING CLASS A2 - 150# CS ASTM A106 Gr.B HIC-resistan
DESIGN P / T 3.5 barg / 75 C	OPER. P / T 0.69 barg / 45.0 C	TEST P (HYDRO) 5.2 barg (1.5xPxST/S=1.0)	PIPE WALL SCH 40 (9.27 mm)
INSULATION N 0 mm	PWHT YES - service (sour/HIC) 595-650 C	NDE 20 % RT of BW; HB <= 200	STRESS CAT 2 - T > 150 C or equipment-connected
P&ID CFU-100-PR-PID-009	FLUID / PHASE P / L Unstabilised naphtha pum	PAINT Epoxy/PU system	SPOOLS / WELDS 7 spools; 25 shop / 2 field

BILL OF MATERIALS (THIS SHEET)

PT	DESCRIPTION	NPS	QTY
1	FLANGE WN RF CL150 B16.5, bore SCH 40, ASTM A105	10"	9
2	PIPE SCH 40, BE, ASTM A106 Gr.B SMLS HIC/SSC (NACE MR0103)	10"	11.1 m
3	ELBOW 90 LR, BW, SCH 40, ASTM A234 WPB	10"	5
4	TEE EQUAL, BW, SCH 40, ASTM A234 WPB	10"	1
5	REDUCER ECC, BW, SCH 40, ASTM A234 WPB	10" x 8"	2
6	FLANGE WN RF CL150 B16.5, bore SCH 40, ASTM A105	8"	2
7	ERECTION: GATE VALVE OS&Y CL150 RF, A216 WCB, trim 8	10"	3
8	TEMPORARY CONE STRAINER (commissioning)	10"	2
9	GASKET SPW 304/graphite, outer ring, CL150 B16.20	10"	10
10	STUD BOLT SET (nuts) A193 B7 / A194 2H, CL150	10"	10
11	PIPE SUPPORTS (std. dwgs): rest x4	-	4

CUT LENGTHS (mm, ex. weld gap) / SPOOLS

PIECE	NPS	L	PIECE	NPS	L	PIECE	NPS	L
SP01-1	10"	1189	SP01-2	10"	2338	SP01-3	10"	903
SP01-4	10"	950	SP02-1	10"	1453	SP02-2	10"	254
SP03-1	10"	335	SP04-1	10"	19	SP04-2	10"	1441
SP05-1	10"	419	SP06-1	10"	335	SP07-1	10"	19
SP07-2	10"	1441						

SPOOL DEVELOPED LENGTHS (transport limit 12.0 x 3.0 x 3.0 m):

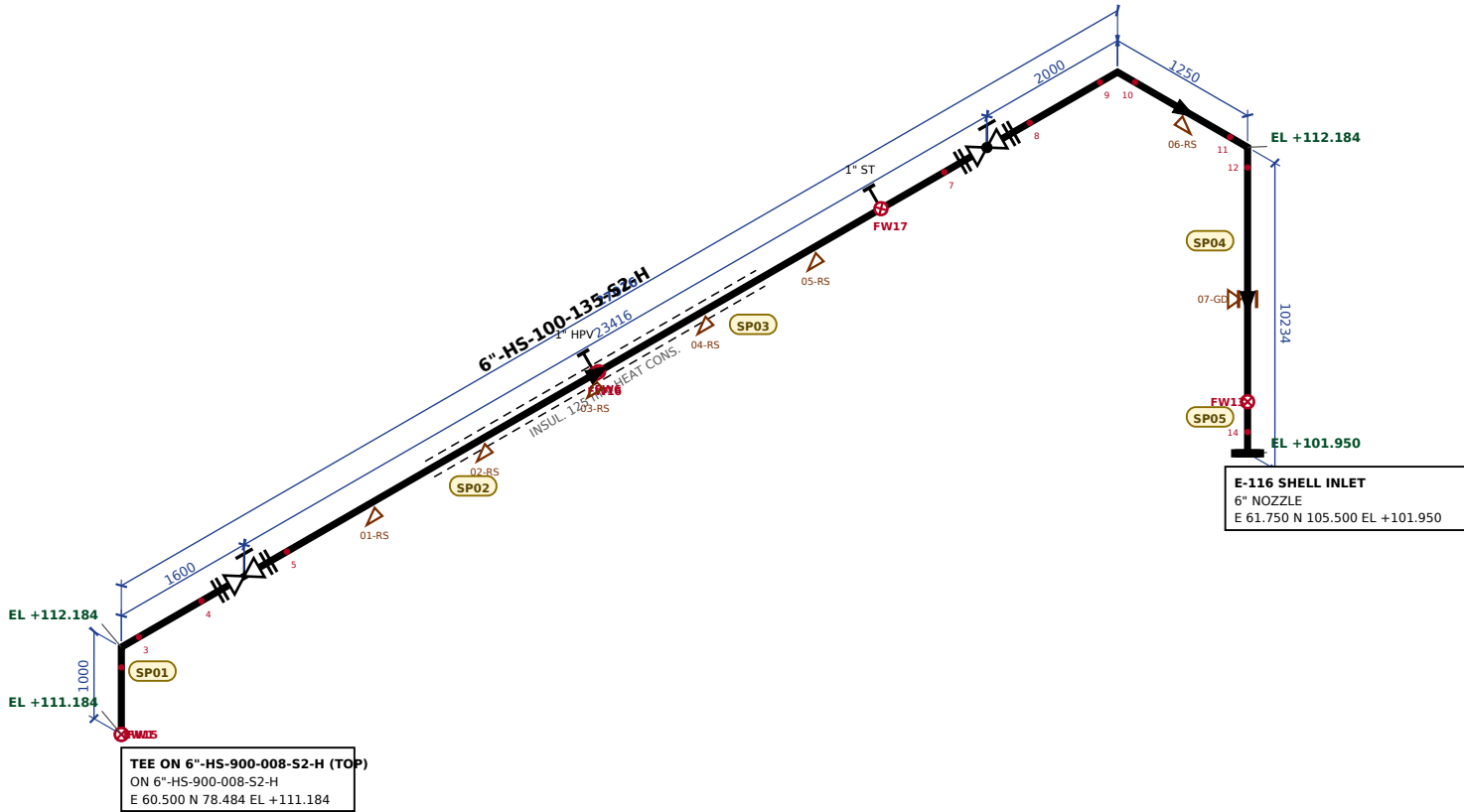
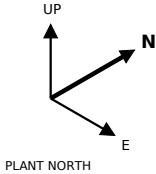
SP01=7.4m SP02=2.6m SP03=0.5m SP04=2.6m SP05=0.5m SP06=0.5m SP07=2.6m

NOTES

- Pipe/fittings/flanges per class A2 (CFU-000-PI-SPC-001); BW fittings ASME B16.9, flanges B16.5/B16.47A RF.
- Hydrotest 5.2 barg per ASME B31.3 345.4 (vents at high points, drains at low points).
- PWHT: YES - service (sour/HIC) 595-650 C. NDE: 20 % RT of BW; HB <= 200.
- Supports: RS rest/shoe, GD guide, ANC anchor, SPR variable spring - tags PS-<area><seq>-nn.
- Welds: dot = shop weld, crossed circle FW = field weld; spools <= 12 m for transport.
- Pump suction: no pockets, header falls to pumps; eccentric reducers 10"x8" FLAT ON TOP; temporary straine
- Pump nozzle 8" assumed (API 610 vendor data pending).
- Vortex breaker in D-102 outlet nozzle (by vessel).
- Common header, equal branch to each pump.

REV	DATE	DESCRIPTION
A	2026-10-02	Issued for review (FEED)
GENERIC REFINERY - GULF COAST 100 KBPSD CRUDE & VACUUM DISTILLATION UNIT		
PIPING ISOMETRIC 10"-P-100-112-A2-N D-102 TO P-104A/B SUCTION		
DISCIPLINE PIPING	STAGE FEED	SCALE NTS
SHEET 1 OF 1		
DRAWN CLAUDE CODE	CHECKED -	APPROVED -
DATE 2026-10-02		
DRAWING NUMBER CFU-100-PI-ISO-011		REV A

ISOMETRIC VIEW - NOT TO SCALE - DIMENSIONS IN mm, ELEVATIONS / COORDINATES IN m (PLANT GRID)



LINE NUMBER 6"-HS-100-135-S2-H	FROM HP steam header	TO E-116	PIPING CLASS S2 - 600# 1.25Cr-1/2Mo ASTM A335 P11
DESIGN P / T 46.0 barg / 430 C	OPER. P / T 41.4 barg / 400 C	TEST P (HYDRO) 81.1 barg (1.5xPxST/S=1.18)	PIPE WALL SCH 40 (7.11 mm)
INSULATION H 125 mm	PWHT YES (P-No.4) 705-745 C	NDE 100 % RT of BW; 100 % MT/PT of fillets;	STRESS CAT 1 - T > 300 C & NPS >= 6; 600# class & T > 200
P&ID CFU-100-PR-PID-010	FLUID / PHASE HS / V HP steam to stabiliser r	PAINT HT silicone/Al (insul.)	SPOOLS / WELDS 5 spools; 11 shop / 6 field

BILL OF MATERIALS (THIS SHEET)

PT	DESCRIPTION	NPS	QTY
1	PIPE SCH 40, BE, ASTM A335 P11 SMLS	6"	36.4 m
2	ELBOW 90 LR, BW, SCH 40, ASTM A234 WP11	6"	3
3	FLANGE WN RF CL600 B16.5, bore SCH 40, ASTM A182 F11	6"	5
4	SOCKOLET 3000# (branch), ASTM A182 F11	1"	2
5	ERECTION: GATE VALVE OS&Y CL600 RF, A217 WC6, trim 8	6"	1
6	GLOBE VALVE CL600 RF, A217 WC6, trim 8	6"	1
7	GASKET SPW 304/graphite, outer ring, CL600 B16.20	6"	5
8	STUD BOLT SET (nuts) A193 B7 / A194 2H, CL600	6"	5
9	PIPE SUPPORTS (std. dwgs): guide x1, rest x6	-	7

CUT LENGTHS (mm, ex. weld gap) / SPOOLS

PIECE	NPS	L	PIECE	NPS	L	PIECE	NPS	L
SP01-1	6"	771	SP01-2	6"	975	SP02-1	6"	11650
SP03-1	6"	10973	SP04-1	6"	1375	SP04-2	6"	793
SP04-3	6"	8750	SP05-1	6"	1138			

SPOOL DEVELOPED LENGTHS (transport limit 12.0 x 3.0 x 3.0 m):

SP01=2.3m SP02=11.8m SP03=11.1m SP04=11.9m SP05=1.3m

NOTES

- Pipe/fittings/flanges per class S2 (CFU-000-PI-SPC-001); BW fittings ASME B16.9, flanges B16.5/B16.47A RF.
- Hydrotest 81.1 barg per ASME B31.3 345.4 (vents at high points, drains at low points).
- PWHT: YES (P-No.4) 705-745 C. NDE: 100 % RT of BW; 100 % MT/PT of fillets; HB <= 225; 100 % PMI.
- Supports: RS rest/shoe, GD guide, ANC anchor, SPR variable spring - tags PS-<area><seq>-nn.
- Welds: dot = shop weld, crossed circle FW = field weld; spools <= 12 m for transport.
- Branch taken from top of HP header; line falls to E-116 - drains into shell (no pocket).
- S2 (P11) - PWHT and 100 % RT; hot bolting check at 430 C.

REV	DATE	DESCRIPTION
A	2026-10-02	Issued for review (FEED)

GENERIC REFINERY - GULF COAST

100 KBPSD CRUDE & VACUUM DISTILLATION UNIT

PIPING ISOMETRIC 6"-HS-100-135-S2-H

HP STEAM HEADER TO E-116

DISCIPLINE	STAGE	SCALE	SHEET
PIPING	FEED	NTS	1 OF 1
DRAWN CLAUDE CODE	CHECKED -	APPROVED -	DATE 2026-10-02
DRAWING NUMBER CFU-100-PI-ISO-012			REV A